Notes on Abstract Algebra and Geometry

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May 22, 2026

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1 Logic

Definition If a statement A leads logically to another statement B we say that ‘if A, then B’ and write $A \to B$. Another way to think about implication is that $A \to B$ can be thought of as ‘if A is true, then B must be true’ (‘B, if A’) or that B is *necessary* for A. This can also be expressed as ‘A *only if* B’.

$A \to B$ implies that A being true is sufficient for B to be true. If the we have both $A \to B$ and $B \to A$, then we can say that A is *necessary and sufficient* for B – which can also be stated as $A \iff B$.

These logical dependences can be summarized in the statement “*If* is sufficient, *Only If* is necessary.”

Definition n-dimensional Affine Space $\mathbb{R}_n$. An ordered n-tuple of real numbers, i.e., a system of $(x_1, x_2, ..., x_n)$ of n real numbers in a definite order, is called a point (n a positive integer). The numbers $x_1, x_2, ..., x_n$ are called the coordinates of the point; in particular x_1 is called its first, x_2 its second, ..., x_n its n-th coordinate. Two points $P = (x_1, x_2, ..., x_n)$ and $Q = (y_1, y_2, ..., y_n)$ are said to be the *same* or to coincide if and only if $x_1 = y_1, x_2 = y_2, ..., x_n = y_n$. The totality of all n-tuples of real numbers is called n-dimensional Affine Space and is denoted by $\mathbb{R}_n$.

Definition The laws of addition:

A1: The Commutative Law of addition: $x + y = y + x$ for all pairs of numbers x and y.

A2: The Associative Law of addition: $(x + y) + z = x + (y + z)$ for any three numbers x, y and z.

A3: The existence of a zero. There is a number, called the zero and denoted by 0, which has the property that

$$x + 0 = x \text{ and } 0 + x = x \text{ for all numbers } x.$$

A4: The existence of negatives: Corresponding to each number x there is a number called the negative of x and written $-x$ which satisfies

$$x + -x = 0 \text{ and } -x + x = 0.$$

Definition Distributive Law

D1: The Distributive Law:

$$\begin{aligned}(x+y)z &= xz+yz\\ x(y+z) &= xy+xz\end{aligned}$$

Definition The laws of multiplication:

M1: The Commutative Law of multiplication : $xy = yx$ for all pairs of numbers x and y.

M2: The Associative Law of multiplication: $(xy)z = x(yz)$ for any three numbers x, y and z.

M3: The existence of a unity. There is a number, called the unity and denoted by 1, which has the property that

$$x1 = x \text{ and } 1x = x \text{ for all numbers } x.$$

M4: The existence of inverses: Corresponding to each number x there is a number called the inverse of x and written x^{-1} which satisfies

$$xx^{-1} = 1 \text{ and } x^{-1}x = 1.$$

2 Mappings

Definition An injective function $f : X \to Y$ (also known as injection, or one-to-one function) is a function f that maps distinct elements of its domain to distinct elements; that is, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. Equivalently, $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$ in the equivalent contrapositive statement.

Definition A surjective function $f : X \to Y$ (also known as surjection, or onto function) is a function f that every element y can be mapped from some element x so that $f(x) = y$. In other words, every element of the function's codomain is the image of at least one element of its domain. It is not required that x be unique; the function f may map one or more elements of the set X to the same element of the set Y.

Definition Homorphisms and Mappings:
Monomorphism: 1-1, Epimorphism: onto, Isomorphism: 1-1 and onto.
Homomorphisms preserve structure (i.e. multiplication and addition in Groups and Rings).
Homomorphisms that have the same object and image space are called Endomorphisms.
And Endomorphism that is also an Isomorphism is called and Automorphism.
In a Group G, the mappings of G into itself, given by $g \to x^{-1}gx$, where x is any fixed element of G, is an Automorphism, known as an Inner Automorphism.

Suppose $f : A \to B$ and $g : B \to C$. Hence $(g \circ f) : A \to C$ exists. Show
(a) If f and g are 1-1 (monomorphisms), then $g \circ f$ is 1-1:
Suppose $(g \circ f)(x) = (g \circ f)(y)$. Then $g(f(x)) = g(f(y))$. Because g is 1-1, $f(x) = f(y)$ and because f is

1-1, $x = y$. Therefore $(g \circ f)(x) = (g \circ f)(y)$, implies $x = y$; hence $g \circ f$ is 1-1.
(b) If f and g are onto (epimorphisms), then $g \circ f$ is onto.
Suppose $c \in C$. Because g is onto, there exists $b \in B$ for which $g(b) = c$. Because f is onto, there exists $a \in A$ for which $f(a) = b$. Thus $(g \circ f)(a) = g(f(a)) = g(b) = c$, Hence $g \circ f$ is onto.

(c) If $g \circ f$ is 1-1, then f is 1-1.
Prove contrapositive, Suppose f is not 1-1., Then there exists distinct elements $x, y \in A$ for which $f(x) = f(y)$. Then $(g \circ f)(x) = g(f(x)) = g(f(y)) = (g \circ f)(y)$. Hence $g \circ f$ is not 1-1.Therefore if $g \circ f$ is 1-1, then f must be 1-1.
(d) If $g \circ f$ is onto, then g is onto.
Prove contrapositive, If $a \in A$, then $(g \circ f)(a) = g(f(a)) \in g(B)$. Hence $(g \circ f)(A) \subseteq g(B)$. Suppose g is not onto. Then $g(B)$ is properly contained in C and so $(g \circ f)(A)$ is properly contained in C; thus $g \circ f$ is not onto. Therefore if $g \circ f$ is onto, then g must be onto.

3 Equivalence Relations

Theorem 3.1 *The equivalence classes theorem*
Suppose in a set S we have a relation R defined between certain pairs of elements, xRy meaning that x and y stand in the given relation R.
Suppose further that R has the following 3 properties:

(1) It is reflexive: i.e. xRx for all $x \in S$.

(2) It is symmetric: i.e. $xRy \Rightarrow yRx$.

(3) It is transitive: i.e. xRy and $yRz \Rightarrow xRz$.

Then R is an equivalence relations: i.e. it divides S into mutually exclusive subsets so that every element of S is in one and only one subset, and so that two elements are in the same subset if and only if they stand in the relation R to one another.

The subsets are called equivalence classes defined by R.

Given any element x in S consider all the elements y such that xRy. These form a subset of S: let us call it A_x. We show first that two elements are in the same subset A_x if and only if they stand in the relation R to one another. Suppose yRz and $y \in A_x$. Then xRy and so xRz since R is transitive. Hence $z \in A_x$. Conversely, if y and z are both in A_x we have xRy and xRz, i.e. yRx by the symmetric property and xRz: thus yRz by transitivity.

For each element x we now have a subset A_x, but these will not all be distinct. We will show that two such are either mutually exclusive or else identical. Suppose A_x and A_y both have an element z, Then xRz and yRz, snf do zRy by the symmetric property; hence xRy by the transitive property. Now take any element w of A_x. Then xRw and since xRy we have yRx, giving yRw. So w is in A_y. Hence we have shown that if A_x and A_y have one element z in common, any element w of A_x is in A_y, i.e, $A_x \subset A_y$. Similarly $A_y \subset A_x$, and so the subsets A_x and A_y are identical.

Thus we have mutually exclusive subsets $A_{x_1}, A_{x_2}, ...$ Finally by the reflexive property any element t is in one of the subsets, viz. A_t.

4 Subgroups and Cosets

Theorem 4.1 *A necessary and sufficient condition for a non-empty subset H to be a subgroup of a group G is that $gh^{-1} \in H$ for all $g, h \in H$.*

(1) Necessary: If H is a subgroup $h^{-1} \in H$ and so $gh^{-1} \in H$.

(2) Sufficient: If g is any element of H we have $gg^{-1} \in H$, i.e, $e \in H$. Hence $eg^{-1} = g^{-1} \in H$, i.e, H contains inverses. Thus if $g, h \in H$, so is h^{-1} and hence $g(h^{-1})^{-1} \in H$, i.e. $gh \in H$.

Given a group G and a subgroup H it is possible to decompose the whole group into subsets 'parallel' to H, two elements being in the same subset if their 'difference' is in H. This concept is better adapted in the product notation where the difference between g and k is expressed as $k^{-1}g$. We therefore decompose group G into subsets such that g and k are in the same subset if $k^{-1}g \in H$. These subsets are known as cosets and we have a true decomposition into mutually exclusive subsets because $k^{-1}g \in H$ sets up an equivalence relation gRk if $k^{-1}g \in H$.

Theorem 4.2 *Equivalence Classes of G relative to H The relation defined by gRk above is an equivalence relation.*

(1) Reflexive: For any $g \in H$ we have gRg because $g^{-1}g = e \in H$ since H is a subgroup.

(2) Symmetric: If gRk then $k^{-1}g \in H$. Hence its inverse is in H, i.e. $g^{-1}k \in H$ and so kRq.

(3) Transitive: If gRk and kRl then $k^{-1}g \in H$ and $l^{-1}k \in H$. Hence $(l^{-1}k)(k^{-1}g) = l^{-1}g \in H$ and so gRl.

These mutual exclusive classes are called left cosets of G relative to H. So if gRk then kRg and $g^{-1}k \in H$ or $k = gh$ for some $h \in H$. Hence the coset containing g is precisely the complex gH, the set $\{gh\}$ for all $h \in H$.

Theorem 4.3 *Left Cosets The left cosets of G relative to H are the complexes gH, $g \in G$. Any left coset may be expressed in this form for any g in it. Any two cosets are either the same of have no element in common.*

We may similarly define *right cosets* of G relative to H as the equivalence classes under the relation given by $gk^{-1} \in H$. They are the complexes Hg.

We normally will use left cosets and therefore will just refer to them as cosets.

5 Invariant Subgroups

Definition H is an invariant subgroup of G if it is a subgroup and if the left and right cosets of H in G coincide; i.e., if

$$gH = Hg \quad \forall g \in G.$$

Invariant subgroups are often called normal subgroups or self-conjugate subgroups.
If the left coset gH is also a right coset, then it contains $g(=eg)$ and so must be the right coset Hg. Hence is every left coset is a right coset (or vice versa) H is invariant. e
Let us show that this condition is sufficient.

Theorem 5.1 *If H is invariant in G then $(gH)(g'H) = gg'H$ for all $g, g' \in G$.*

Proof

$$\begin{aligned} gHg'H &= g(Hg')H = g(g'H)H \text{ since } Hg' = g'H \text{ as H is invariant,} \\ &= gg'H^2 = gg'H \text{ since } H^2 = H \text{ as } H \text{ is a subgroup.} \end{aligned}$$

Not that is would have been sufficient to prove that

$$(gH)(g'H) \subseteq gg'H,$$

Theorem 5.2 *A subgroup H is an invariant subgroup of G is and only if $g^{-1}Hg = H$ for all $g \in G$.*

6 Conjugacy

Definition The elements x and $g^{-1}xg$, where x and g are elements of a group G, are called conjugate elements in $G:$ in other words x and y are conjugate if and only if there exists and element $g \in G$ such that $y = g^-1xg$.

Theorem 6.1 *The relation of conjugacy between elements of any group G is an equivalence relations.*

Reflexive. Since $x = e^{-1}x$, x is conjugate to itself.

Symmetric. If $y = g^{-1}xg$ we have $x = gyg^{-1} = (g^{-1})^{-1}yg^{-1}$.

Transitive. If $y = g^{-1}xg$ and $z = h^{-1}yh$ we have

$$z = h^{-1}g^{-1}xgh = (gh)^{-1}x(gh).$$

It follows that the relation divides the elements of G into mutually exclusive classes; these are called *conjugacy classes in G.*

Theorem 6.2 *If H is a subgroup of G then $g^{-1}Hg$ is a subgroup for any element $g \in G$.*

Proof For

$$(g^{-1}Hg)^2 = (g^{-1}Hgg^{-1}Hg) = (g^{-1}HHg) = (g^{-1}Hg) \text{ since } H^2 = H$$

as H is a subgroup. Also

$$(g^{-1}Hg)^{-1} = g^{-1}H^{-1}g = g^{-1}Hg \text{ since } H^-1 = H.$$

We say that H and $g^{-1}Hg$ are *conjugate subgroups* in G.

Theorem 6.3 *The relation of conjugacy between subgroups of G is an equivalence relation.*

Theorem 6.4 *H is an invariant subgroup of G if the conjugacy class of subgroups of G which contains H consists of just H itself: i.e., if the only subgroup conjugate to H is H.*

7 Quotient groups

Theorem 7.1 *If H is an invariant subgroup of G, the cosets of H in G form a group under the group product given by*

$$(gH)(g'H) = gg'H.$$

The product is well defined and unique, and is also a coset.
It is associative, since

$$((gH)(g'H))(g''H) = gg'g''H = (gH)((g'H)(g''H)).$$

The coset $H = (eH)$ is a unity since

$$(gH)(eH) = geH = gH \text{ and } (eH)(gH) = egH = gH.$$

The coset gH has an inverse $g^{-1}H$, since

$$(gH)(g^{-1}H) = gg^{-1}H = eH = H \text{ and } (g^{-1}H)(gH) = H$$

similarly.
The quotient group formed by the cosets of H and G is called the *Quotient Group of H in G*, or sometimes the *Factor Group of G relative to H*. sectionHomomorphisms As we have already hinted, the subjects of invariant subgroups and homomorphisms of groups are closely connectedl there is in fact a 1-1 correspondence between subgroups of a given group G and the essentially different homomorphism which have G ax the object space, so that the study of either tops is similar to that of the other.

Suppose that H is an invariant subgroup of G, so that the quotient group G/H is defined. Consider the mapping $\theta : G \to G/H$ defined by $g\theta = gH$, the coset containing g, which is of course an element of G/H.

Theorem 7.2 *The mapping $\theta : G \to G/H$ defined by $g\theta = gH$ is a homomorphism.*

Proof

$$\begin{aligned}(gg')\theta &= gg'H = gg'HH \\ &= gHg'H \\ &= (g\theta)(g'\theta)\end{aligned}$$

Corollary 7.3 *The kernel θ is H and the image is G/H, so that θ is an epimorphism.*

The homomorphism θ is called the Natural Homomorphism or the Canonical Homomorphism associated with H. It is essential that H be an invariant subgroup, for otherwise G/H is not defined and we cannot speak of the product of the cosets $(gH)(g'H)$.

8 Quotient Group Associated with a Homomorphism

Any homomorphism $\theta : G \to G'$ is closely related to the natural homomorphsim associated with an invariant subgroup of G.

Theorem 8.1 *The kernel of $\theta : G \to G'$ is an invariant subgroup of G.*

If h is in the kernel H and g is any element of G, we have

$$\begin{aligned}(g^{-1}hg)\theta &= (g^{-1}\theta)(h\theta)(g\theta) \\ &= (g\theta)^{-1})f(g\theta), \text{ where } f \text{ is the neutral element of } G' \\ &= (g\theta)^{-1}(g\theta) = f\end{aligned}$$

Theorem 8.2 *The mapping defined by $gH \to g\theta$ is a well-defined mapping of G/H onto $G\theta$.*

If g_1 and g_2 are in the same coset, then $g_1\theta = g_2\theta$. If g_1 and g_2 are in the same coset, then $g_2^{-1}g_1 \in H$. Hence $(g_2\theta)^{-1}(g_1\theta) = f$, the neutral element in G', since H is in the kernel of θ. Therefore $g_2\theta = g_1\theta$.
Also any element of $G\theta$ is the image of some $g \in G$ and so the mapping is onto $G\theta$, although not necessarily onto G'.

Theorem 8.3 *The mapping defined by $gH \to g\theta$ is an isomorphism between G/H and $G\theta$.*

We have that

$$\begin{aligned}(g_1H)(g_2H) &= g_1g_2H, \\ (g_1\theta)(g_2\theta) &= (g_1g_2)\theta,\end{aligned}$$

because θ is a homomorphism and $g_1g_2H \to (g_1g_2)\theta$.
Since this mapping is onto $G\theta$, it is also an isomorphism provided it is 1-1, i.e., provided the kernel is the neutral element of G/H. Suppose $gH \to f$. Then $g\theta \to f$ and so $g \in H$, so that $gH = H$, the neutral element of G/H.
Notive that the sets of elements of G which map into the particular elements of $G\theta$ are the cosets of G relative to the kernel H.
We have proved that to any invariant subgroup H of G there is a natural homomorphism from G onto G/H, and conversely that any homomorphism of G is isomorphic to a natural homomorphism associated with its kernel. Therefore there is a 1-1 correspondence between the invariant subgroups and the essential different homomorphisms of G. By essentially different homomorphisms we mean those having different kernels: two homomorphisms having the same kernel are both equivalent to the same natural homomorphism abd therefore exhibit the same structure.
The image space G' is largely irrelevant : the subgroup $G\theta$ of G' is isomorphic to G/H but G' may be larger. Also there is nor reason why $G\theta$ should be invariant in G'.

Theorem 8.4 *First Isomorphism Theorem: Let ϕ be a group homomorphism from G tp G'. Then the mapping from $G/Ker(\phi)$ to $G\phi$ given by $gKer(\phi) \to g\phi$ is an isomorphism.*

Example Consider the homomorphism $\phi : \mathbb{Z} \to \mathbb{Z}_4$ such that $x\phi = x \mod 4$. In this case let G be the group of integers (under addition)

$$G = \{..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...\}$$

and

$$G' = \mathbb{Z}_4 = \{0, 1, 2, 3\}.$$

Also

$$Ker(\phi) = \{... - 8, , -4, 0, 4, 8, ...\} = \langle 4 \rangle \text{ (the cyclic subgroup generated by 4)}$$

So $G\phi = \{0, 1, 2, 3\}$. Note that $G\phi$ doesn't have to be the entire group G'. In this example it is the case that $G\phi = G'$. Then

$$G/Ker(\phi) = \{0 + \langle 4 \rangle, 1 + \langle 4 \rangle, 2 + \langle 4 \rangle, 3 + \langle 4 \rangle\} \text{ (four cosets)}$$

Notice that we can create a mapping $\overline{\phi}$ that mapps $G/Ker(\phi)$ into $G\phi$ as follows:

$$\overline{\phi} : G/Ker(\phi) \to g\phi,$$

where

$$\begin{aligned}
\overline{\phi}(0 + \langle 4\rangle) &= (0)\phi = 0 \\
\overline{\phi}(1 + \langle 4\rangle) &= (1)\phi = 1 \\
\overline{\phi}(2 + \langle 4\rangle) &= (2)\phi = 2 \\
\overline{\phi}(3 + \langle 4\rangle) &= (3)\phi = 3
\end{aligned}$$

Theorem 8.5 *The mapping $\phi : \mathbb{Z} \to \mathbb{Z}_n$ defined by $m\phi = m \mod n$ is a homomorphism with kernel $\langle n\rangle$. By the First Isomorphism Theorem, $\mathbb{Z}/\langle n\rangle \cong \mathbb{Z}_n$.*

If ϕ is a homomorphism from a finite group G to G', then $\|G\phi\|$ divides $\|G\}$ and $\|G'\|$. This follows from the First Isomorphism Theorem and the fact that $G\phi$ is a subgroup of G' and Lagrange's Theorem.

Lemma 8.6 *The innter automorphism θ_x given by $g \to x^{-1}gx$ is the identity automorphism if and on if $x \in C$ where C is the center.*

If $x \in C$, $x^{-1}gx = x^{-1}xg = g$ and so $\theta_x = I$, the identity.
If $\theta_x = I, x^{-1}gx = g\forall g$ and so $x \in C$.

Lemma 8.7 *The mapping $x \to \theta_x$ of G into the group of automorphisms of G is a homomorphism*

We already have that $\theta_{xy} = \theta_x\theta_y$ and so the result follows.
By Lemma(8.6) the kernel of this homomorphism is the center C. The image is the subgroup of inner automorphisms. Therefore we obtain

Theorem 8.8 *The group of inner automorphisms of a group $G \cong G/C$, where C is the center of G.*

9 Relationship between Normal Subgroups and Their Images

Consider a group G that has a normal subgroup N and H and that $H \subseteq N$.

Theorem 9.1 *Because $N \supseteq H$, both N and H are normal subgroups of G. Thus G/N is the homomorphic image of G/H under the natural homomorphsim $\pi : G/H \to G/N$ defined by $(xG)\pi = xN$.*

Proof 1) Define a function $\phi : G \to G/N$ by

$$x\phi = xN.$$

This is the standard canonical projection map. It is a surjective group homomorphism with $ker(\phi) = N$.
2) We want to construct a map $\overline{\phi} : G/H \to G/N$. Define $\overline{\phi}$ by:

$$(xH)\overline{\phi} = x\phi = xN.$$

3) We must show that if $xH = yH$, then $(xH)\overline{\phi} = (yH)\overline{\phi}$.

$$\begin{aligned}
xH &= yH \to y^{-1}x \in H. \\
H &\subseteq N \text{ given}
\end{aligned}$$

$$
\begin{aligned}
y^{-1}x &\in N \\
xN &= N \\
(xH)\overline{\phi} = (yH)\overline{\phi}
\end{aligned}
$$

The map is well-defined because $H \subseteq ker(\phi)$.

4) $\overline{\phi}$ is a Homomorphism:

$$(xH)(yH)\overline{\phi} = (xyH)\overline{\phi} = xyN = (xN)(yN) = (xH)\overline{\phi}(yH)\overline{\phi}.$$

5) $\overline{\phi}$ is Surjective: Let $yN \in G/N$. Since $y \in G$, we can form the coset $yH \in G/H$:

$$(yH)\overline{\phi} = yN.$$

Every element in the co-domain has a pre-image
6) Apply the First Isomorphism Theorem By the First Isomorphism Theorem, the domain modulo the kernel is isomorphic to the image

$$(G/H)/ker(\overline{\phi}) \cong Im(\overline{\phi}) = G/N.$$

Since G/N is isomorphic to a quotient group of G/H, it is by definition a homomorphic image of G/H. ∎

10 Fundamental Theorem of Algebra

Theorem 10.1 *(Nested Intervals Therorem): Suppose that*

$$I_n = \{x : a_n \leq x \leq b_n\}, \quad n = 1, 2, ...,$$

is a sequence of closed intervals such that $I_{n+1} \subset I_n$, for each n. If $\lim_{n\to\infty}(b_n - a_n) = 0$, then there is one and only one number x_0 which is in every I_n.

Proof By hypothesis, we have $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$. Since $a_n < b_n$ for every n, the sequence $\{a_n\}$ is nondecreasing and bounded from above by b_1. Similarly, the sequence b_n is nonincreasing and bounded from below by a_1. Using Axiom C, we conclude that there are numbers x_0 and x_0' such that $a_n \to x_0, a_n \leq x_0$, and $b_n \to x_0', b_n \geq x_0'$ as $n \to \infty$. Using the fact that $b_n - a_n \to 0$ as $n \to \infty$, we find that $x_0 = x_0'$ and

$$a_n \leq x_0 \leq b_n$$

for every n. Thus x_0 is in every I_n.
To show that x_0 is the only number in every I_n, suppose there is another number x_1 also in every I_n. Define $\epsilon = |x_1 - x_0|$. Now, since $a_n \to x_0$ and $b_n \to x_0$, there is an integer N_1, such that $a_{N_2} > x_0 - \epsilon$; also there is an integer N_2 such that $b_{N_2} < x_0 + \epsilon$. These inequalities simply that I_n cannot contain x_1 for any n beyond N_1 and N_2. Thus x_1 is not in every I_n. ∎

Theorem 10.2 *Suppose f is continuous on $[a, b], x_n \in [a, b]$ for each n, and $x_n \to x_0$. Then $x_0 \in [a, b]$ and $f(x_n) \to f(x_0)$.*

Proof Since $x_n \geq a$ for each n, it follows from the theorem on the limit of inequalities for sequences that $x_0 \geq a$. Likewise $x_0 \leq b$ so $x_0 \in [a, b]$. If $a < x_0 < b$, then $f(x_n) \to f(x_0)$ on account of the composite function theorem. The same conclusion holds if $x_0 = a$ or b.

Theorem 10.3 *(Intermediate-value theorem): Suppose f is continuous on $[a, b], c \in \mathbb{R}, f(a) < c,$ and $f(b) > c$. Then there is at least one number x_0 on $[a, b]$ such that $f(x_0) = c$.*

There may be more than one such point!

Proof Define $a_1 = 1$ and $b_1 = b$. Then observe that $f((a_1 + b_1)/2)$ is either equal to c, greater than c, or less than c. If it equals c, then choose $x_0 = (a_1 + b_1)/2$ and the result is proved. If $f((a_1 + b_1)/2) > c$, then define $a_2 = a_1$ and $b_2 = (a_1 + b_1)/2$. If $f((a_1 + b_1)/2 < c$, then define $a_2 = (a_1 + b_1)/2$ and $b_2 = b_1$.
In each of the last two cases we have $f(a_2) < c$ and $f(b_2) > c$. Again compute $f((a_2 + b_2)/2)$. If this value equals c, the result is proved. If $f((a_2 + b_2)/2) > c$, set $a_3 = a_2$ and $b_3 = (a_2 + b_2)/2$. If $f((a_2 + b_2)/2 < c$, then define $a_3 = (a_2 + b_2)/2$ and $b_3 = b_2$.
Continuing in this way, we either find a solution in a finite number of steps or we find a sequence $\{[a_n, b_n]\}$ of closed intervals each of which is one of the two halves of the preceding one, and for which we have

$$b_n - a_n = 2^{1-n}(b_1 - a_1),$$

$$f(a_n) < c, \quad f(b_n) > c, \text{ for each } n,$$

From the Nested Intervals Theorem it follows that there is a unique point x_0 in all these intervals and

$$\lim_{n\to\infty} a_n = x_0 \text{ and } \lim_{n\to\infty} b_n = x_0.$$

From the theorem of continuous function on a sequence of points in a closed interval, we conclude that

$$f(a_n) \to f(x_0) \text{ and } f(b_n) \to f(x_0).$$

From the inequalities about and the limit of inequalities it follows that $f(x_0) \leq c$ and $f(x_0) \geq c$ so that $f(x_0) = c$.

Definition A polynomial over a field F is a *rational* and *integral* expression of the form

$$f(x) = p_0 x^n + p_1 x^{n-1} + ... + p_{n-1} x + p_n, \tag{1}$$

where the coefficients $p_i \in F, i = 1, 2, \ldots, n$ and *rational* means free from fractional powers and *integral* means never in the denominator of a fraction. A polynomial function is a function which can be written in the form of Eq. (1).

If $p_0 = 1$ we say the polynomial is in *monic* form.

The roots $\{a_1, a_2, \ldots a_n\}$ of a polynomial are the values of x such that $f(x) = 0$. The roots are in general complex numbers. According to a theorem of Viète, the relationship between the roots and the coefficients for a monic polynomial is given by the

$$\begin{aligned} f(x) &= x^n + p_1 x^{n-1} + \ldots + p_{n-1} x + p_n \\ &= (x - a_1)(x - a_2)(x - a_3) \ldots (x - a_n). \end{aligned}$$

Expanding the product on the right-hand side we find the explicit relationships:

$$\begin{aligned}
p_1 &= -(a_1 + a_2 + \ldots + a_n) \\
p_2 &= (a_1a_2 + a_1a_3 + \ldots + a_2a_3 + \ldots + a_{n-1}a_n) \\
p_3 &= -(a_1a_2a_3 + a_1a_2a_4 + \ldots + a_2a_3a_4 + \ldots + a_{n-2}a_{n-1}a_n) \\
&\vdots \\
p_n &= -(1)^n(a_1a_2a_3a_4 \ldots a_{n-1}a_n)
\end{aligned}$$

These relationships are encapsulated in the following theorem

Theorem 10.4 *In every algebraic equation, the coefficient of whose highest term is unity, the coefficient p_1 of the second term with its sign changed is equal to the sum of the roots.*

The coefficient p_2 of the third terms is equal to the sum of the products of the roots taken two by two.

The coefficient of the fourth term with its sign changed is equal to the sum of the products of the roots taken three by three; and so on, the signs of the coefficients being taken alternately negative and positive, and the number of roots multiplied together in each term of the corresponding function of the roots increasing by unity till finally that function is reached with consists of the produces of the n roots.

Definition The Elementary Symmetric functions $c_1, c_2, c_3, \ldots c_n$ are defined as functions of n indeterminants $x_1, x_2, x_3, \ldots, x_n$ as follows

$$\begin{array}{lll}
s(x_1) & = x_1 + x_2 + \ldots + x_n & = c_1 \\
s(x_1x_2) & = x_1x_2 + x_1x_3 + \ldots + x_2x_3 + \ldots + x_{n-1}x_n & = c_2 \\
s(x_1x_2, x_3) & = x_1x_2x_3 + x_1x_2x_4 + \ldots + x_2x_3x_4 + \ldots + x_{n-2}x_{n-1}x_n & = c_3 \\
\cdots & \cdots\cdots\cdots & \cdots \\
s(x_1x_2 \ldots x_n) & = x_1x_2x_3x_4 \ldots x_{n-1}x_n & = c_n
\end{array} \qquad (2)$$

Theorem 10.5 *The sums of the similare powers of the roots of an equation can be expressed in terms of the coefficients (the elementary symmetric functions).*

Proof Consider the polynomial

$$\begin{aligned}
f(x) &= x^n + p_1x^{n-1} + p_2x^{n-2} \ldots + p_{n-1}x + p_n, \\
&= (x - a_1)(x - a_2)(x - a_3) \ldots (x - a_n) = 0.
\end{aligned}$$

Define the following terms for the sums of the powers of the roots:

$$s_i = \sum_{j=1}^{n} \alpha_j^i, \quad i = 0, 1, 2, ...$$

Given that $f(x)$ is a polynomial we can write the derivative of $f(x)$ as

$$\begin{aligned}
f'(x) &= \frac{f(x)}{x - a_1} + \frac{f(x)}{x - a_1} + \ldots + \frac{f(x)}{x - a_1} \\
&= nx^{n-1} + (n-1)p_1x^{n-2} + (n-2)x^{n-3} + \ldots + 2p_{n-2}x + p_{n-1}
\end{aligned}$$

Using the method of Synthetic Division to divide $f(x)$ by the monomial $(x - a)$ we write out the sequence of computations as follows;

$$\begin{array}{rr|r|r|r}
\frac{f(x)}{x-a} & = x^{n-1} + a & x^{n-2} + a^2 & x^{n-3} + a^3 & x^{n-4} + \ldots + a^{n-2} \\
 & +p_1 & +ap_1 & +a^2 p_1 & +a^{n-2} p_1 \\
 & & +p_2 & +ap_2 & +a^{n-3} p_2 \\
 & & & +p_3 & + \ldots \\
 & & & & +ap_{n-2} \\
 & & & & +p_{n-1}
\end{array}$$

By replacing a by each of the quantities $a_1, a_2, ...a_n$ in succession and adding and using $s_p = \sum a^p$ we obtain the following expression for the derivative of $f(x)$:

$$\begin{array}{rr|r|r|r}
f'(x) & = x^{n-1} + s_1 & x^{n-2} + s_2 & x^{n-3} + s_3 & x^{n-4} + \ldots + s_{n-2} \\
 & +np_1 & +s_1 p_1 & +s_2 p_1 & +s_{n-2} p_1 \\
 & & +np_2 & +sp_2 & +s_{n-3} p_2 \\
 & & & +np_3 & + \ldots \\
 & & & & +sp_{n-2} \\
 & & & & +np_{n-1}
\end{array}$$

Therefore we obtain the following relationships between the s_i's and the coefficients p_i:

$$\begin{aligned}
s_1 + p &= 0 \\
s_2 + p_1 s_1 + 2p_2 &= 0 \\
s_3 + p_1 s_2 + p_2 s_1 + 3p_3 &= 0 \\
s_4 + p_1 s_3 + p_2 s_3 + p_3 s_1 + 4p_4 &= 0 \\
\ldots\ldots\ldots\ldots & \\
s_{n-1} + p_1 s_{n-2} + p_2 s_{n-3} + \ldots + p_{n-2} s_1 + (n-1) p_{n-1} &= 0
\end{aligned}$$

We now proceed to extend our results to the sums of all positive powers of the roots, $s_n, s_{n+1}, \ldots, s_m$ by multiplying the polynomial by x^{m-n} and using the $s_0 = n$, we obtain

$$x^{m-n} f(x) = x^m + p_1^{m-1} + p_2 x^{m-2} + \ldots + p_n x^{m-n},$$

By replacing x by the roots $a_1, a_2, \ldots, a_n$ in succession and adding we have,

$$s_m + p_1 s_{m-1} + p_2 s_{m-2} + \ldots + p_n s_{m-n} = 0.$$

Letting m have the values $m = n, n+1, n+2, \ldots$ successively and noting that $s_0 = n$, we obtain the following equations

$$\begin{aligned}
s_n + p_1 s_{n-1} + p_2 s_{n-2} + \ldots + np_n &= 0 \\
s_{n+1} + p_1 s_n + p_2 s_{n-1} + \ldots + p_n s_1 &= 0 \\
s_{n+2} + p_1 s_{n+1} + p_2 s_n + \ldots + p_n s_2 &= 0 \\
\ldots\ldots\ldots & \quad 0
\end{aligned}$$

By transforming the equation $f(x) = 0$ into one whose roots are the reciprocals of $a_1, a_2, a_3, \ldots, a_n$, and applying the above formulas we can also express all the negative powers of the roots.

Theorem 10.6 *Every rational symmetric function of $x_1, x_2, \ldots, x_n$ is rationally expressible in terms of the elementary symmetric functions.*

Proof

Theorem 10.7 (Theorems related to Symmetric Functions) *The sum of the exponents of all the roots in any term of any symmetric function of the roots is equal to the sum of the suffices in each term of the corresponding value in terms of the coefficients.*

The sum of the exponents is of course the same for every term of the symmetric function, and may be called the degree in all the roots of that function. In other words the suffix for each coefficient is equal to the degree in the roots of the corresponding function of the roots; hence in any product of any powers of the coefficients the sum of the suffices must equal to the degree in all the roots of the corresponding function of the roots.

Theorem 10.8 *When an equation is written with binomial coefficients, the expression in terms of the coefficients for any symmetric function of the roots, which is a function of their differences only, is such that the algebraic sum of the numerical factors of all the terms in it is equal to zero.*

Theorem 10.9 (Division Algorithm for Polynomials) *If F is any field, and $a(x) \neq 0$ and $b(x)$ are any two polynomials over F, then we can find polynomials $q(x)$ and $r(x)$ such that*

$$b(x) = q(x)a(x) + r(x),$$

where $r(x)$ is either zero or has a degree less than that of $a(x)$.

Theorem 10.10 *In $F[x]$, any two polynomials a and b have a greatest common divisor (GCD) d satisfying (i) $d|a$ and $d|b$, (ii) $c|a$ and $c|b$ imply that $c|d$. Moreover, (iii) d is a linear combination $d = sa + tb$ of a and b.*

The procedure for determining the GCD of $a(x)$ and $b(x)$ is given in Algorithm (1):

Algorithm 1 Polynomial Euclidean Algorithm

```
procedure POLYGCD(a(x), b(x))
    while b(x) ≠ 0 do
        r(x) ← a(x) mod b(x)
        a(x) ← b(x)
        b(x) ← r(x)
    end while
    return a(x)
end procedure
```

Theorem 10.11 (Eisenstein's Irreducibility Criterion) *For a given prime p, let $a(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots a_n$ be a polynomial with integral coefficients, such that $a_n \neq 0(\mod p)$, $\quad a_{n-1} = a_{n-2} = a_0 = 0(\mod p), a_0 \neq 0(\mod p^2)$. Then $a(x)$ is irreducible over the field of rational.*

Theorem 10.12 (Euler-Gauss: Fundamental Theorem of Algebra) *Every polynomial p(x) of positive degree with complex coefficients has a complex root.*

11 Determinants and Their Properties

Consider a set of vectors $V = \{\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n\}$ in $\mathbb{R}^n$. The Determinant $D(\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n)$ has the following properties

i) Invariance property: $D(\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n)$ is to remain unchanged if some $\mathfrak{a}_i$ is replaced by $\mathfrak{a}_i + \mathfrak{a}_j \quad (i \neq j)$;

ii) Homogeneity property: $D(\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n)$ is to go over into $\lambda D(\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n)$ if $\mathfrak{a}_i$ is replaced by $\lambda \mathfrak{a}_i$;

iii) the Normalization: $D(e_1, e_2, ..., e_n) = 1$.

The determinant is the unique function of vectors $\{\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n\}$ that satisfy the above properties [1].

Conventions: We adopt the Range and Summation Conventions:

Definition Range Convention. *When a small Latin suffix (superscript or subscript) occurs unrepeated in a term, it is understood to take all the values* $1, 2, ..., n$, *where* n *is the number of dimensions of the space.*

Definition Summation Convention. *When a small Latin suffix is repeated in a term, summation with respect to that suffix is understood, the range of summation being* $1, 2, ..., n$.

Kronecker delta:

$$\delta_r^s = \begin{cases} 1 & \text{if } s = r \\ 0 & \text{if } r \neq s \end{cases}$$

Also, the another form of the Kronecker delta is δ_{rs} which satisfies the same cases as above.

Permutation Symbol: Consider a set σ of $\sigma_1, \sigma_2, ..., \sigma_n$ numbers taken from the set $\{1, 2, ..., n\}$. Let

$$\epsilon_{\sigma_1 \sigma_2 ... \sigma_n} = \begin{cases} +1 & \text{if } \sigma \text{ is an even permutation of } \{1, 2, ..., n\} \\ -1 & \text{if } \sigma \text{ is an odd permutation of } \{1, 2, ..., n\} \\ 0 & \text{if } \sigma \text{ has any duplicates} \end{cases}$$

Let the S_n be the set of all permutation of the numbers $\{1, 2, ..., n\}$. In the case of n-by-n matrix A we obtain the complete development of the determinant of A as follows:

$$\det |A| = \sum_{\sigma \in S_n} \epsilon_{\sigma_1 \sigma_2 ... \sigma_n} a_{1\sigma_1} a_{2\sigma_2} ... a_{n\sigma_n} = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} \tag{3}$$

If the rows are permuted by $\rho \in S_n$, we find

$$\epsilon_{\rho_1 \rho_2 ... \rho_n} \det |A| = \sum_{\sigma \in S_n} \epsilon_{\sigma_1 \sigma_2 ... \sigma_n} a_{\rho_1 \sigma_1} a_{\rho_2 \sigma_2} ... a_{\rho_n \sigma_n} \tag{4}$$

Cofactor of an element a_{rs},

$$M_{rs} = \text{cofactor}(a_{rs}) = \sum_{\sigma \in S_n, \ \sigma_r = s} \epsilon_{\sigma_1 \sigma_2 ... \sigma_n} a_{1\sigma_1} a_{2\sigma_2} ... a_{r-1\sigma_{r-1}} a_{r+1\sigma_{r+1}} .... a_{n\sigma_n} \tag{5}$$

Relation between cofactors and minors obtained by taking the determinant obtained from A by striking out row r and column s from A. We denote such minor determinants by A_{rs} :

$$M_{rs} = (-1)^{r+s} A_{rs}$$

For every r, $1 \le r \le n$ we can write $\det |A|$ in terms of a cofactor expansion

$$\det |A| = \sum_{s=1}^{n} a_{rs}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{rs}\, A_{rs} \tag{6}$$

Also, if there is a mismatch between the row indices between the matrix element and the cofactor or minor, i.e.,

$$\sum_{s=1}^{n} a_{ks}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{ks}\, A_{rs} = 0, \text{ for } \quad k \neq r \tag{7}$$

The case $k \neq r$ corresponds to a determinant with two identical rows and hence vanishes because the rows are linearly dependent. Putting both of the above equations together we find

$$\sum_{s=1}^{n} a_{ks}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{ks}\, A_{rs} = \delta_{kr} \det |A| \tag{8}$$

Classical Adjoint and Inverse Matrix We define the Classical Adjoint, also known as the Adjugate, of the matrix A as

$$\mathbf{adj}(A) = M^{\dagger},$$

where $M^{\dagger}$ is the transpose of the cofactor matrix, M. Using Eq.(8), the inverse of A is given by

$$A^{-1} = \frac{1}{|A|}\mathbf{adj}.(A) = \frac{1}{|A|} M^{\dagger}.$$

Since $A^{-1}A = I$ we see that the determinant of A^{-1} is $|A|^{-1}$. Therefore the determinant of the Classical Adjoint matrix is

$$|\mathbf{adj}(A)| = |A|^{n} |A|^{-1} = |A|^{n-1},$$

for an n-by-n matrix A.

Left Inverse and Right Inverse Suppose we have a left inverse X and a right inverse Y of a matrix A. This means that

$$XA = AY = I.$$

Therefore we have

$$\begin{aligned} XAY &= X, \quad \text{and} \\ XAY &= Y. \end{aligned}$$

Therefore

$$X = XAY = Y, \text{ or } X = Y.$$

Therefore left and right inverses of a non-singular matrix must be equal.

The inverse of a Symmetric Matrix is Symmetric A symmetric matrix is defined by $A = A^\dagger$. The inverse of A satisfies

$$A^{-1}A = I$$

Therefore, by taking the transpose of both sides we have

$$A^T(A^{-1})^\dagger = A(A^{-1})^\dagger = I.$$

Therefore

$$A^{-1}A(A^{-1})^\dagger = A^{-1}I = A^{-1},$$

or

$$(A^{-1})^\dagger = A^{-1}.$$

Therefore the inverse of a symmetric matrix is also symmetric.

Special Cases – Inverting a Triangular Matrix

Theorem 11.1 (Upper Triangular Matrices) *Let U be an upper triangular matrix. If the inverse U^{-1} exists, then it is upper triangular.*

Proof Recall the $(n-1)\times(n-1)$ cofactor matrix, M_{rs} that results from omitting row r and column s of $U = (u_{ij})$. When it exists,

$$U^{-1} = \frac{1}{|U|}\mathbf{adj}(U),$$

so it is enough to prove that the $n \times n$ matrix $(|C_{rs}|)$ of cofactor determinants whose transpose is $(|C_{rs}|)^\dagger$ is the adjugate, is lower triangular.

In the case $r < s$, every element below the diagonal of the matrix C_{rs} is also below the diagonal of U, so must be equal to 0.

Hence C_{rs} is upper triangula, with the determinant equal to the product of its diagonal elements.

However, $s - r$ of these diagonal elements are $u_{i+1,j}$ for $i = r, \ldots, s-1$. These elements are from below the diagonal of U, and so are equal to 0.

Hence $r < s$ implies that $|C_{rs}| = 0$, so the $n \times n$ matrix $(|C_{rs}|)$ of cofactor determinants is indeed lower triangular. Hence the adjugate $(|C_{rs}|)^\dagger$ is upper triangular. Hence the inverse of and upper triangular matrix is upper triangular. By similar reasoning we can show that the inverse of a lower triangular matrix is lower triangula. ∎

Reciprocal Vector Space in n-Dimensions Consider a set of n vectors $\{\mathfrak{a}'_1, \mathfrak{a}'_2, ..., \mathfrak{a}'_n\}$, given by

$$\mathfrak{a}'_j = \frac{1}{|A|}\{M_{j1}, M_{j2}, \ldots, M_{jn}\}.$$

We will refer to this set of vectors as being reciprocal to the set of vectors forming the columns of A. From the definition of inverse of matrix A we immediately have

$$\mathfrak{a}_k \cdot \mathfrak{a}'_j = \delta_{kj}.$$

Also the determinant of the n-by-n matrix formed from the set of $\mathfrak{a}'_j, j = 1, 2, ..., n$ vectors arranged in columns

$$A' = \begin{bmatrix} \mathfrak{a}'_1 & \mathfrak{a}'_2 & \dots & \mathfrak{a}'_n \end{bmatrix} \tag{9}$$

had a determinant equal to

$$\left|A'\right| = |A|^{-n}|\det(M) = |A|^{-n}||A|^{n-1} = |A|^{-1}. \tag{10}$$

Therefore the volume of the set of reciprocal vectors volume$\{\mathfrak{a}'_1, \mathfrak{a}'_2, \dots, \mathfrak{a}'_n\} = \dfrac{1}{|A|} = \dfrac{1}{V}$ where $V = |A|$ is the volume of the set of the vectors $\{\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_n\}$. This is the n-dimensional Euclidean space version of Reciprocal Vectors in 3-dimensional Euclidean space.

Schur's complement: Suppose p, q are nonnegative integers, and suppose A, B, C, D are respectively p-by-p, p-by-q, q-by-p, and q-by-q matrices of complex numbers. Let

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \tag{11}$$

so that M is a $(p + q)$-by-$(p + q)$ matrix.

If D is invertible, then the Schur complement of the block D of the matrix M is the p-by-p matrix defined by

$$M/D := A - BD^{-1}C. \tag{12}$$

and

$$\det M = \det D \det(A - BD^{-1}C).$$

If A is invertible, the Schur complement of the block A of the matrix M is the q-by-q matrix defined by

$$M/A := D - CA^{-1}B. \tag{13}$$

and

$$\det M = \det A \det(D - CA^{-1}B).$$

In the case that A or D is singular, substituting a generalized inverse for the inverses on M/A and M/D yields the generalized Schur complement.

Laplace's General Expansion Theorem: Divide the set of integers $1, 2, ..., n$ into two complementary sets $\{\alpha_1, \alpha_2, ..., \alpha_p\}$ and $\{\beta_1, \beta_2, ..., \beta_q\}$ where $n = p + q$ and

$$\alpha_1 < \alpha_2 < ... < \alpha_p \text{ and } \beta_1 < \beta_2 < ... < \beta_q.$$

These two sets remained fixed throughout the following:

$$\det A = \sum_{\substack{\rho_1, \rho_2, \dots, \rho_p \\ \rho_1 < \rho_2 < \dots < \rho_p}} (-1)^{(\alpha_1+\alpha_2+\ldots+\alpha_p+\rho_1+\rho_2\ldots+\rho_p)} \begin{vmatrix} a_{\alpha_1\rho_1} & a_{\alpha_1\rho_2} & \cdots & a_{\alpha_1\rho_p} \\ a_{\alpha_2\rho_1} & a_{\alpha_2\rho_2} & \cdots & a_{\alpha_2\rho_p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\alpha_i\rho_1} & a_{\alpha_i\rho_2} & \cdots & a_{\alpha_i\rho_p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\alpha_p\rho_1} & a_{\alpha_p\rho_2} & \cdots & a_{\alpha_p\rho_p} \end{vmatrix} \begin{vmatrix} a_{\beta_1\sigma_1} & a_{\beta_1\sigma_2} & \cdots & a_{\beta_1\sigma_q} \\ a_{\beta_2\sigma_1} & a_{\beta_2\sigma_2} & \cdots & a_{\beta_2\sigma_q} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\beta_i\sigma_1} & a_{\beta_i\sigma_2} & \cdots & a_{\beta_i\sigma_q} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\beta_q\sigma_1} & a_{\beta_q\sigma_2} & \cdots & a_{\beta_q\sigma_q} \end{vmatrix} \tag{14}$$

The sets $\{\alpha_1, \alpha_2, ..., \alpha_p\}$ and $\{\beta_1, \beta_2, ..., \beta_q\}$ were defined to be fixed. The summation is taken over all p-tuples $(\rho_1, \rho_2, ..., \rho_p)$ from among the numbers $1, 2, ..., n$ for which $\rho_1 < \rho_2 < ... < \rho_p$ and where

$$\sigma_1 < \sigma_2 < ... < \sigma_q$$

are the remaining q integers.

12 Cauchy-Binet Formula

Application 1: **Example Partitioned Matrix and the Cauchy-Binet Formula**

$$M = \left[\begin{array}{cccc|cccc} 0 & 0 & \dots & 0 & b_{11} & b_{12} & \dots & b_{1n} \\ 0 & 0 & \dots & 0 & b_{21} & b_{22} & \dots & b_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & b_{m1} & b_{m2} & \dots & b_{mn} \\ \hline c_{11} & c_{12} & \dots & c_{1m} & 1 & 0 & \dots & 0 \\ c_{21} & c_{22} & \dots & c_{2m} & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ c_{n1} & c_{n2} & \dots & c_{nm} & 0 & 0 & \dots & 1 \end{array}\right] \tag{15}$$

In the above M is an $(m+n)$-by-$(m+n)$ matrix which is partitioned into: A an m-by-m matrix of 0's, B an m-by-n matrix, C an n-by-m matrix and D is the n-by-n Identity matrix. Then using the determinant in the Schur's Complement form we have

$$\det M = \det D \det(A - BD^{-1}C) = \det(-BC) = (-1)^m \det(BC).$$

We can obtain a similar result by row operations. Denote the $(m+i)$ row by $\mathfrak{a}_i$. Adding a multiple of row i to row j does not change the value of $\det M$. We successively subtract the $\sum_{i=1}^{n} b_{k\,i}\mathfrak{a}_i$ from the k-th row of M, for $k = 1, 2, ..., m$ changing M into a different matrix with the same determinant, viz.,

$$\tilde{M} = \left[\begin{array}{cccc|cccc} -d_{11} & -d_{12} & \dots & -d_{1m} & 0 & 0 & \dots & 0 \\ -d_{21} & -d_{22} & \dots & -d_{2m} & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ -d_{m1} & -d_{m2} & \dots & -d_{mm} & 0 & 0 & \dots & 0 \\ \hline c_{11} & c_{12} & \dots & c_{1m} & 1 & 0 & \dots & 0 \\ c_{21} & c_{22} & \dots & c_{2m} & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ c_{n1} & c_{n2} & \dots & c_{nm} & 0 & 0 & \dots & 1 \end{array}\right], \tag{16}$$

where $d_{kj} = \displaystyle\sum_{i=1}^{n} b_{ki}c_{ij}$, for $k, j = 1, ..., m$. Since $\det M = \det \tilde{M}$ we have another demonstration of

$$\det M = (-1)^m \det(BC) \tag{17}$$

where B and C are the matrices corresponding to b_{kj} and c_{ij} respectively.

Another representation of $\det M$ can be obtained by applying Laplace's Generalized Theorem successively. First swap the bottom n rows with the top m rows (by moving each of the bottom n rows up m positions and changing the sign of $\det M$ accordingly

$$\det M = (-1)^{m^2} \det \left[\begin{array}{cccc|cccc} c_{11} & c_{12} & \dots & c_{1m} & 1 & 0 & \dots & 0 \\ c_{21} & c_{22} & \dots & c_{2m} & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ c_{n1} & c_{n2} & \dots & c_{nm} & 0 & 0 & \dots & 1 \\ \hline 0 & 0 & \dots & 0 & b_{11} & b_{12} & \dots & b_{1n} \\ 0 & 0 & \dots & 0 & b_{21} & b_{22} & \dots & b_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & b_{m1} & b_{m2} & \dots & b_{mn} \end{array}\right] \tag{18}$$

Using Laplace's Generalized Theorem the first time across the bottom m rows and taking advantage of the m-by-m block of zeros in the lower left hand corner to drop some of the minors in the expansion we have only a sum over the m-by-m minors in the b_{ij} elements starting in columns $m+1$ through $m+n$ of M. After forming any one of these minors we have to select m columns to remove to form the algebraic complement to use in the Laplace Theorem. The n-by-n complementary minor consists of the n-by-m c_{ij} matrix elements with an additional $n-m$ columns which remain in from the former Identity matrix in the upper right hand side of the working matrix.

We now apply Laplace's Generalized Theorem a second time, now expanding over the first m columns of the complementary n-by$n$ minor. We must again sum over the choice of m rows from the first n rows. The complementary minor to this second set of minors is obtained from the original n-by-n Identity matrix, I_n, with m columns removed by forming the first set of minors in the b_{ij} and removing the m rows to form the second set of minors. We are left with an $n-m$-by-$n-m$ matrix from I_n. If the rows that were struck out do not match exactly the columns that were struck out then there will appear in the the reduced matrix rows and columns with all zero entries and the corresponding determinant vanishes. So in the second application of Laplace's Generalized theorem there is only one term surviving in the sum, for any given choice of m numbers $1 \leq \rho_1 \leq \rho_2 \leq ... \leq \rho_m \leq n$, the only term that survive is

$$\begin{vmatrix} c_{1\rho_1} & c_{1\rho_2} & \dots & c_{1\rho_m} \\ c_{2\rho_1} & c_{2\rho_2} & \dots & c_{2\rho_m} \\ \dots & \dots & \dots & \dots \\ c_{m\rho_1} & c_{m\rho_2} & \dots & c_{m\rho_m} \end{vmatrix} \begin{vmatrix} b_{1\rho_1} & b_{1\rho_2} & \dots & b_{1\rho_m} \\ b_{2\rho_1} & b_{2\rho_2} & \dots & b_{2\rho_m} \\ \dots & \dots & \dots & \dots \\ b_{m\rho_1} & b_{m\rho_2} & \dots & b_{m\rho_m} \end{vmatrix} \tag{19}$$

Putting both results together and using the fact that both methods give an expression for $\det M$ we obtain the Cauchy-Binet Formula for matrices B (n-by-m) and C (m-by-n) with $m \leq n$:

$$\det(BC) = \sum_{\substack{\rho_1,\rho_2,\dots,\rho_m=1 \\ \rho_1<\rho_2<\dots<\rho_m}}^{n} \begin{vmatrix} c_{1\rho_1} & c_{1\rho_2} & \dots & c_{1\rho_m} \\ c_{2\rho_1} & c_{2\rho_2} & \dots & c_{2\rho_m} \\ \dots & \dots & \dots & \dots \\ c_{m\rho_1} & c_{m\rho_2} & \dots & c_{m\rho_m} \end{vmatrix} \begin{vmatrix} b_{1\rho_1} & b_{1\rho_2} & \dots & b_{1\rho_m} \\ b_{2\rho_1} & b_{2\rho_2} & \dots & b_{2\rho_m} \\ \dots & \dots & \dots & \dots \\ b_{m\rho_1} & b_{m\rho_2} & \dots & b_{m\rho_m} \end{vmatrix} \tag{20}$$

1: For rectangular matrices, when the number of columns exceeds the number of rows. The overall determinant is formed summing the product of all the sub-determinants be formed from one array (*by taking a number of columns equal to the number of rows*) multiplied by the corresponding determinants from the other array.

2: When the number of rows exceeds the number of columns, the resulting determinant vanishes. It will be observed that the resulting determinant is the same as would arise if an appropriate number of zero-columns were added to each of the given arrays and then forming the determinant of the product.

Application 2: **Alternate Derivation of the Cauchy-Binet Formula**
Let A be an m-by-n matrix and let B be an n-by-m matrix. Let $1 \leq j_1, j_2, ..., j_m \leq n$. Let $A_{j_1 j_2 ... j_m}$ denote the m-by-m matrix consisting of the m columns $j_1, j_2, ..., j_m$ of A. Let $B_{j_1 j_2 ... j_m}$ denote the m-by-m matrix consisting of the m rows $j_1, j_2, ..., j_m$ of B. Let $\{k_1, k_2, ..., k_m\}$ be an ordered m-tuple of integers and let $\{j_1, j_2, ..., j_m\}$ be the same set of integers but arranged into non-decreasing order:

$$\text{j}_1 \leq j_2 \leq ... \leq j_m.$$

Then we have the relationships that

$$\begin{aligned} \det\left(B_{k_1 k_2 ... k_m}\right) &= \epsilon_{k_1 k_2 ... k_m} \det\left(B_{j_1 j_2 ... j_m}\right), \\ \det\left(B_{j_1 j_2 ... j_m}\right) &= \epsilon_{k_1 k_2 ... k_m} \det\left(B_{k_1 k_2 ... k_m}\right) \text{ no implied sum on } k_i's. \end{aligned}$$

Now from the definition of determinant for the m-by-m matrix AB we have

$$\begin{aligned} \det(AB) &= \sum_{1 \leq l_1, ... l_m \leq m} \epsilon_{l_1 l_2 ... l_m} \left(\sum_{k=1}^{n} a_{1k} b_{kl_1}\right) ... \left(\sum_{k=1}^{n} a_{mk} b_{kl_m}\right) \\ &= \sum_{1 \leq k_1, ... k_m \leq n} \left(a_{1k_1} ... a_{mk_m}\right) \sum_{1 \leq l_1, ... l_m \leq m} \epsilon_{l_1 l_2 ... l_m} b_{k_1 l_1} ... b_{k_m l_m} \\ &= \sum_{1 \leq k_1, ... k_m \leq n} \left(a_{1k_1} ... a_{mk_m}\right) \det\left(B_{k_1 ... k_m}\right) \\ &= \sum_{1 \leq k_1, ... k_m \leq n} \left(a_{1k_1} ... a_{mk_m} \epsilon_{k_1 ... k_m}\right) \det\left(B_{j_1 ... j_m}\right) \\ &= \sum_{1 \leq k_1, ... k_m \leq n} \left(\epsilon_{k_1 ... k_m} \, a_{1k_1} ... a_{mk_m}\right) \det\left(B_{j_1 ... j_m}\right) \\ &= \sum_{1 \leq j_1 \leq j_2 \leq ... \leq j_m \leq n} \det\left(A_{j_1 ... j_m}\right) \det\left(B_{j_1 ... j_m}\right) \end{aligned}$$

If any two j's are equal then $\det\left(A_{j_1 ... j_m}\right) = 0$
In the case of A and $B = A^T$ we immediately have

$$\det(AA^T) = \sum_{1 \leq j_1 \leq j_2 \leq ... \leq j_m \leq n} \left[\det\left(A_{j_1 ... j_m}\right)\right]^2$$

13 Substitutions and Almost Symmetric Functions

Substition refers to the set of permutations of n elements which form the Symmetric Group.

Theorem 13.1 *All the substitutions of $x_1, x_2, \ldots, x_n$ which leave unaltered a rational function $\phi(x_1, x_2, \ldots, x_n$ form a group G.*

Let ϕ_s denote the function obtained by applying to ϕ the substitusion s. If a and b are two substitutions which leave ϕ unaltered, then $\phi_a = \phi, \phi_b = \phi$. Then

$$(\phi_a)_b = (\phi)_b = \phi_b = \phi, \text{ or } \phi_{ab} = \phi$$

Hence the product of ab is one of the substitutions which leave ϕ unaltered and the set G has the group property.

The group G is called the *group of the function* ϕ, while ϕ is said to *belong to the group* G

Theorem 13.2 *Every substitution can be expressed as a product of transpositions in various ways.*

Theorem 13.3 *Of the various decompositions of a given substitution s into a product of transpositions, all contain an even number of transpositions (an even substitution), or all constain an odd number of transpositions (an odd substitution).*

Corollary 13.4 *The totality of even substitutions on n letters forms a group, called the Alternating Group on n letters.*

Consider n elements $x = \{x_1, x_2, x_3, \ldots, x_n\}$. Consider the rational function

$$\begin{array}{rlll} \phi(x) = \prod(x_i - x_j) = & (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) & \ldots\ldots & (x_1 - x_n) \\ & (x_2 - x_3)(x_2 - x_4) & \ldots\ldots & (x_2 - x_n) \\ & (x_3 - x_4) & \ldots\ldots & (x_3 - x_n) \\ & & \ldots\ldots\ldots & \\ & & & (x_{n-1} - x_n) \end{array}$$

consisting of the product of all the difference of the elements.

$\phi(x)$ is related to the Vandemonde determinant

$$\phi(x) = V(x) = \begin{vmatrix} 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & x_2^3 & \ldots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & x_3^3 & \ldots & x_2^{n-1} \\ \vdots & & & & & \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^{n-1} \end{vmatrix}$$

The square of $V(x)$ is a well- known symmetric function, the discriminant $\triangle$. Therefore $V(x)$ has two values equal numerically with opposites signs, viz. $\sqrt{\triangle}$ and $-\sqrt{\triangle}$. $V(x)$ is an example of an alternating function. Any transposition alters the sign of $V(x)$. For example, the transposition (x_1x_2) affects only the terms in the first and second lines of the product, and replaces them by

$$\begin{array}{rll} (x_2 - x_1)(x_2 - x_3)(x_2 - x_4) & \ldots\ldots & (x_2 - x_n) \\ (x_1 - x_3)(x_1 - x_4) & \ldots\ldots & (x_1 - x_n). \end{array}$$

Hence, if s is the product of an even number of transpositions, it leaves $v(x)$ unaltered; if s is the product of an odd number of transpositions, it changes $V(x)$ into $-V(x)$.

Conjugate values of Multiple-valued Function and Conjugate Groups

Theorem 13.5 (Lagrange) *The order of a subgroup is a divisor of the order of the group.*

Proof Consider a group G of N and a subgroup H composed of the substitutions

$$h_1 = I, h2, h3, \ldots, h_P. \tag{21}$$

If G constains no further substitutions, then $N = P$ and the theorem is true. Now let G contain a substitution g_2 not in H. Then G also must contain

$$g_2, h_2g_2, h_3g_2, \ldots, h_Pg_2. \tag{22}$$

This set are all distinct and all different from the substitutions (21), since $h_\alpha g_2 = h_\beta$ requires that $g_2 = h_\alpha^{-1}h_\beta =$ a substition of H contrary to the assumption that g_2 is distinct from the elements of H. Hence the substitutions combined now give $2P$ distinct substitutions of G. If there are no other substitutions in G, then $N = 2P$ and the theorem is true. If there are any other distrinct substitutions g_3 in G not in (21) or (22), then G contains

$$g_3, h_2g_3, h_3g_3, \ldots, h_Pg_3. \tag{23}$$

Moreover, these substitutions are all different from (21) and (22), since $h_\alpha g_3 = h_\beta g_2$ requires that $g_3 = h_\alpha^{-1}h_\beta g_2$ and shall belong to (refHL2), contrary to assumption. We now have $3P$ distinct substitutions of G. Either $N = 3P$ or else G contains a substitution g_4 not in the sets (21), (23), (23). In the latter case G contains the products

$$g_4, h_2g_4, h_2g_4, \ldots, h_Pg_4, \tag{24}$$

all of which are distrinct and all different from the sets (21), (23), (23), so that we have $4P$ distinct substitutions. Proceeding in this way, we finally reach a last set of P substitutions

$$g_nu, h_2g_nu, h_3g_nu, \ldots, h_Pg_\nu, \tag{25}$$

since the order of$H$ is finite. Hence $N = \nu P$.

Definition The number ν is called the *index* of the subgroup H under G and is denoted by $\nu = [G : H]$.

Corollary 13.6 *The order of any group H of substitutionis on n letters is a divisor of $n!$. H is a subgroup of the symmetric group G_n on n letters.*

Theorem 13.7 *The period of any substitution contained in a group G of order N is a divisor of N.*

Corollary 13.8 (Applied to the Symmetric Group) *The order of any substitution group on a set of n elements is an exact divisor of $N = n!$, the quotient being the number of distinct value of the corresponding multiple-valued function.*

Let us arrange the N substitutions of$G$ in a rectangular array using the substitutions of a subgroup H in the first row:

$$\begin{array}{lllll} h_1 = I & h_2 & h_3 & \ldots & h_P \\ g_2 & h_2g_2 & h_3g_2 & \ldots & h_Pg_2 \\ g_3 & h_2g_3 & h_3g_3 & \ldots & h_Pg_3 \\ \vdots & \ldots & \ldots & \ldots & \vdots \\ g_\nu & h_2g_\nu & h_3g_\nu & \ldots & h_Pg_\nu \end{array} \tag{26}$$

The $g_1 = I, g_2, g_3, \ldots, g_\nu$ multiply the elements of H on the right hand side forming right cosets. Similarly, a rectangular array for the substitiions could be formed by using left costs.

Theorem 13.9 *If ψ is a rational funciton of $x_1, x_2, \ldots, x_n$ belonging to a subgroup H of index ν, then ψ is ν-valued under the full set of substitutions of G.*

Proof Apply to ψ all the N substitutions of G arranged in a rectangular array as in (26). All the substitutions belonging to the same row give the same answer since

$$\psi_{h_i g_\alpha} = (\psi_{h_i})_{g_\alpha} = (\psi)_{g_\alpha} = \psi_{g_\alpha}.$$

Hence there result at most ν values. But, if

$$\psi_{g_\alpha} = \psi_{g_\beta}$$

then $\psi_{g_\alpha g_beta^{-1}} = \psi$, so that $g_\alpha g_\beta^{-1} h_i \in H$, leaving ψ unaltered. Hence $g_\alpha = h_i g_\alpha$, contrary to the assumption made in forming the rectangular array.

Regarding the orders of a group H and a larger group G containing H as a subgroup; that is to say, the order r of H is an exact divisor of the order G which contains H as a subgroup, the quotient m being the number of distinct values of a multiple-valued function unaltered by the substitutions of G which are obtained by the substutions of H.

Definition The ν distinct functions $\psi, \psi_{g_2}, \psi_{g_3}, \ldots, \psi_{g_\nu}$ are called **conjugate values** of ψ under G.

Theorem 13.10 *The ρ distinct values which a rational function $\psi(x_1, \ldots, x_n)$ takes when operated on by all $n!$ substitutions are the roots of an equation of degree ρ whose coefficients are ration functions of the elementary symmetric functions.*

Formation of Functions of a Given Group

We define the Galois Function having $n!$ values for all the substiutionts of the symmetric group

$$\psi_1 = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \ldots \alpha_n x_n,$$

in which $\alpha_1, \alpha_2, \ldots, \alpha_n$ are n distinct arbitrary constants. This function is called the Galois Function. If we use all the P substitutions of H to form the functions $\psi_1, \psi_2, \ldots, \psi_P$ by the substitutions of H anysymmetric funciton of these values will be unaltered by the substitutions of H. In particular $\phi_1 = (y - \psi_1)(y - \psi_2) \ldots (y - \psi_P)$ will be unaltered by the substutions of H, and altered to a different value by the substitutions not in H. The function ϕ_1 is not therefore unaltered by the substitutions of any wider group that contains H as a subgroup. Noting the forms of the expression of the sums of powers of the roots in terms of the coefficients, we see that instead of the coefficients of the powers of y in ϕ_1 we may take sums of powers of $\psi_1, \psi_2, \ldots, \psi_P$ up to the P^{th} and deduce that one at least of such P sums of powers will provide a function unlatered by the substitutions of H and altered by any other substitution of the symmetric group.

Theorem 13.11 *Every integral symmetric function of the distinct values of any integral multiple-valued function of n elements is a symmetric function of the elements themselves.*

Proof Let $F(\phi_1, \phi_2, \ldots, \phi_\rho)$ be any rational integral symmetric function of the ρ-values. Any substitution S whatever affecting the elements applied to these ρ-values either leaves any function unchanged or replaces it by one of the others. No two of the resulting values can be equal, for if $\phi_i S$ were equal to $\phi_j S$ it would follow, by applying the substitution S^{-1}, that $\phi_i = \phi_j$ which is contrary to the assumptions. Consequently the ρ values of ϕ are reproduced by S in some order or other. The symmetric function F therefore remains unchanged by any substitutiont, and is consequently a symmetric element of the elements themselves.

Corollary 13.12 *The ρ distinct values of an integral multiple-valued function are roots of an equation whose coefficients are integral symmetric functions of the elements themselves.*

Theorem 13.13 (Lagrange Theorem on functions belonging to the same group) *Two rational functions belonging to the same group are rationally expressed each in terms of the other.*

Proof Let ϕ_1 and ψ_1 be two functions belonging to the same group

$$H = \{I, S_2, S_3, \ldots, S_r\},$$

of order r and degree n, each of these functions having ρ distinct values, where $r * \rho = N$. Any substitution not contained in H will convert ϕ_1 into another of its values, say ϕ_k and at the same time convert ψ_1 into ψ_k. By operating all possible substitutions, ρ pairs of values $\phi_1, \psi_1 : \phi_2, \psi_2 : \ldots \phi_\rho, \psi_\rho$ are obtained. Now, in the first place, the rational function

$$\sum \phi^i \psi^j = \phi_1^i \psi_1^j + \phi_2^i \psi_2^j + \ldots + \phi_k^i \psi_k^j + \ldots + \phi_\rho^i \psi_\rho^j \tag{27}$$

is clearly a symmetric function of the elements, for it appears by the same reasoning as before that any substitution acting on this function will merely permute the terms leavng the sum unaltered. Therefore $\sum phi^i \psi^j$ is a symmetric function of the elements.

If we now take $j = 1$ and assign to i all the values $0, 1, 2, \ldots, \rho - 1$ in succession, we obtain the following ρ equations linear in $\psi_1, \psi_2, \ldots, \psi_\rho$:

$$\begin{array}{rcrcccrc}
\psi_1 & + & \psi_2 & + & \ldots & + & \psi_\rho = & T_0 \\
\phi_1\psi_1 & + & \phi_2\psi_2 & + & \ldots & + & \phi_\rho\psi_\rho = & T_1 \\
\phi_1^2\psi_1 & + & \phi_2^2\psi_2 & + & \ldots & + & \phi_\rho^2\psi_\rho = & T_2 \\
\ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots = & \ldots \\
\phi_1^{\rho-1}\psi_1 & + & \phi_2^{\rho-1}\psi_2 & + & \ldots & + & \phi_\rho^{\rho-1}\psi_\rho = & T_{\rho-1}
\end{array} \tag{28}$$

where $T_0, T_1, \ldots, T_{\rho-1}$ are all symmetric functions of $x_1, x_2, \ldots, x_n$.

We can write the above as a linear system

$$\begin{bmatrix}
1 & 1 & 1 & 1 & \ldots & 1 \\
\phi_1 & \phi_2 & \phi_3 & \phi_4 & \ldots & \phi_\rho \\
\phi_1^2 & \phi_2^2 & \phi_3^2 & \phi_4^2 & \ldots & \phi_\rho^2 \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
\phi_1^{\rho-1} & \phi_2^{\rho-1} & \phi_3^{\rho-1} & \phi_4^{\rho-1} & \ldots & \phi_\rho^{\rho-1}
\end{bmatrix}
\begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \vdots \\ \psi_\rho \end{bmatrix}
=
\begin{bmatrix} T_0 \\ T_1 \\ \vdots \\ \vdots \\ T_{\rho-1} \end{bmatrix} \tag{29}$$

Linear systems such as Eq.(29) can be solved using Cramer's rule. Alternatively, in order to understand better the relatonship between ϕ_i and ψ_i let us consider a method based on transforming non-homogenous linear systems into sets of corresponding homogeneous linear systems in each of the unknown parameters. To this effect consider the linear system [2].

$$\begin{array}{rcrcrcl}
x & + & y & + & z & = & A_0 \\
\alpha x & + & \beta y & + & \gamma z & = & A_1 \\
\alpha^2 x & + & \beta^2 y & + & \gamma^2 z & = & A_2
\end{array} \tag{30}$$

By adding a fourth equation to the above we can isolate one of the variables, say y, and determine the functional dependence of y on β and the symmetric functions of the coefficients. Adding $y = y$ to the above system and converting it into a system of homogenous equations by introducing a 4th unknown w we obtain the system

$$\begin{array}{ccccccc} 0 & + & y & + & 0 & = & y \\ x & + & y & + & z & = & A_0 \\ \alpha x & + & \beta y & + & \gamma z & = & A_1 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & = & A_2 \end{array} \tag{31}$$

The above system can be converted into a Homogeneous System by moving the final column to the left-hand side of the equal sign.

$$\begin{array}{ccccccccc} 0 & + & y & + & 0 & - & y & = & 0 \\ x & + & y & + & z & - & A_0 & = & 0 \\ \alpha x & + & \beta y & + & \gamma z & - & A_1 & = & 0 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & - & A_2 & = & 0 \end{array} \tag{32}$$

Writing the above system using a matrix of coefficients and introducing the 4th undetermined variable we find

$$\begin{bmatrix} 0 & 1 & 0 & -y \\ 1 & 1 & 1 & -A_0 \\ \alpha & \beta & \gamma & -A_1 \\ \alpha^2 & \beta^2 & \gamma^2 & -A_2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{33}$$

In order for this homogenous system to be consistent with Eq.(30) the determinant of the matrix of coefficients must vanish.

$$\begin{vmatrix} 0 & 1 & 0 & -y \\ 1 & 1 & 1 & -A_0 \\ \alpha & \beta & \gamma & -A_1 \\ \alpha^2 & \beta^2 & \gamma^2 & -A_2 \end{vmatrix} \tag{34}$$

Since the determinant vanishes we can factor out the -1 from the last column and write the consistency condition as

$$\begin{vmatrix} 0 & 1 & 0 & y \\ 1 & 1 & 1 & A_0 \\ \alpha & \beta & \gamma & A_1 \\ \alpha^2 & \beta^2 & \gamma^2 & A_2 \end{vmatrix} = 0. \tag{35}$$

Multiplying the above determinant by the non-singular determinant

$$\begin{vmatrix} 1 & 1 & 1 & 0 \\ \alpha & \beta & \gamma & 0 \\ \alpha^2 & \beta^2 & \gamma^2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \tag{36}$$

we find

$$\begin{vmatrix} 0 & 1 & 0 & y \\ 1 & 1 & 1 & A_0 \\ \alpha & \beta & \gamma & A_1 \\ \alpha^2 & \beta^2 & \gamma^2 & A_2 \end{vmatrix} \begin{vmatrix} 1 & \alpha & \alpha^2 & 0 \\ 1 & \beta & \beta^2 & 0 \\ 1 & \gamma & \gamma^2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & \beta & \beta^2 & y \\ s_0 & s_1 & s_2 & A_0 \\ s_1 & s_2 & s_3 & A_1 \\ s_2 & s_3 & s_4 & A_2 \end{vmatrix} = 0, \tag{37}$$

where

$$s_i = \alpha^i + \beta^i + \gamma^i$$

and the s_i are symmetric functions in $\alpha, \beta,$ and γ.

Since the s_i's are sums of powers of α, β, γ and are symmetric functions, they can be expressed rationally in terms of the elementary symmetric functions. By expanding the final determinant in Eq. (37) along the top row and setting the expansion equal to zero to satisfy the consistency requirement we see that we have y expressed as a quadratic function of β along with the sums of powers of the three quantities α, β, γ.
Returning to the original problem Eq. (28), we can similarly express the solution for ψ_j in terms of ϕ_j in the form of a determinant:

$$\begin{vmatrix} 1 & \phi_j & \phi_j^2 & \dots & \phi_j^{\rho-1} & \psi_j \\ s_0 & s_1 & s_2 & \dots & s_{\rho-1} & T_0 \\ s_1 & s_2 & s_3 & \dots & s_\rho & T_1 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ s_{\rho-1} & s_\rho & s_{\rho+2} & \dots & s_{2\rho-2} & T_{\rho-1} \end{vmatrix} = 0, \tag{38}$$

where $T_0, T_1, \dots, T_{\rho-1}$ are all symmetric functions of $x_1, x_2, \dots, x_n$ and $s_k = \phi_1^k + \phi_2^k + \phi_3^k + \dots + \phi_\rho^k$.

The solution of these equations takes the form

$$\triangle(\phi_1, \phi_2, \phi_3, \dots, \phi_\rho)\psi_j = A_0\phi_j^{\rho-1} + A_1\phi_j^{\rho-2} + \dots + A_{\rho-1}, \tag{39}$$

where $\triangle$ is the product of the square of the differences of all the ϕ's (the square of the Vandermonde determinant) and $A_0, A_1, \dots, A_{\rho-1}$ are all symmetric in $x_1, x_2, \dots, x_n$.

Therefore ψ_j can be rationally expressed as a rational function of ϕ_j and the elementary symmetric functions of $x_1, x_2, \dots, x_n$ (the coefficients of the polynomial that has $x_1, x_2, \dots, x_n$ as roots).

Lagrange Interpolation Formula:
Given a set of $k+1$ nodes $\{x_0, x_1, \dots, x_k\}$, which must all be distinct, $x_j \neq x_m$ for indices $j \neq m$, the *Lagrange basis* for polynomials of degree $\leq k$ for those nodes is the set of polynomials $\{l_0(x), l_1(x), \dots, l_k(x)\}$ each of degree k which take values $l_j(x_m) = 0$ if $m \neq j$ and $l_j(x_j) = 1$. Using the Kronecker delta this can be written $l_k(x_m) = \delta_{jm}$. Each basis polynomial can be explicitly described by the product:

$$\begin{aligned} l_j(x) &= \frac{(x-x_0)}{(x_j-x_0)}\frac{(x-x_1)}{(x_j-x_1)}\cdots\frac{(x-x_{j-1})}{(x_j-x_{j-1})}\frac{(x-x_{j+1})}{(x_j-x_{j+1})}\cdots\frac{(x-x_k)}{(x_j-x_k)} \\ &= \prod_{\substack{0\leq m\leq k \\ m\neq j}} \frac{x-x_m}{x_j-x_m} \end{aligned}$$

Notice that the numerator $\prod_{m\neq j}(x-x_m)$ has k roots at the nodes $\{x_m\}_{m\neq k}$ while the denominator $\prod_{m\neq j}(x_j-x_m)$ scales the resulting polynomial so that $l_j(x_j) = 1$.

The Lagrange interpolating polynomial for those nodes through the corresponding values $\{y_0, y_1, \dots, y_k\}$ is the linear combination

$$L(x) = \sum_{j=0}^{k} y_j l_j(x),$$

Each basis polynomial has degree k, so the sum $L(x)$ has degree $\leq k$, and it interpolates the data because

$$L(x_m) = \sum_{j=0}^{k} y_j l_j(x_m) = \sum_{j=0}^{k} y_j \delta_{mj} = y_m,$$

Theorem 13.14 (Extension of Lagrange's Theorem to Non Symmetric Functions) *If two rational functions of any set of variables are such that one remains unchanged by all the substitutions of the group to which the other belongs, the first is expressible by means of the second in the form of an integral polynomial whose coefficients are rational symmetric functions of the variables.*

Another statement of Lagrange's Theorem:

If a rational function $\phi(x_1, x_2, \ldots, x_n)$ remains unaltered by all the substitutions which leae another rational function $\psi(x_1, x_2, \ldots, x_n)$ unaltered, then ϕ is a rational function of ψ and $c_1, c_2, \ldots, c_n$. Note that the group, G to which ϕ belongs contains the group, H to which ψ belongs, i.e. $H \subseteq G$. G is the wider group. Also phi is therefore the more symmetric function – a wider set of substitutions keep it unaltered.

Proof

$$\begin{array}{cccc|c|c} I & h_2 & \ldots & h_P & \psi = \psi_1 & \phi = \phi_1 \\ g_2 & h_2 g_2 & \ldots & h_P g_2 & \psi_{g_2} = \psi_2 & \phi_{g_2} = \phi_2 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ g_\nu & h_2 g_\nu & \ldots & h_P g_\nu & \psi_{g_\nu} = \psi_\nu & \phi_{g_\nu} = \phi_\nu \end{array} \tag{40}$$

The $\psi_1, \psi_2, \ldots, \psi_\nu$ are all distinct; but$\phi_1, \phi_2, ...\phi_n u$ need not be distinct since ϕ belongs to a group G which may be bigger than H. Under any substitution s on $x_1, x_2, \ldots, x_n$, the functions $\psi_1, \psi_2, \ldots, \psi_\nu$ are merely permuted. Moreover, if s replaces ψ_i by ψ_j, it replaces ϕ_i by ϕ_j. Set (using the Lagrange Interpolation formula)

$$\begin{aligned} g(t) &= (t - \psi_1)(t - \psi_2) \ldots, (t - \psi_\nu), \\ \lambda(t) &= g(t) \left(\frac{\phi_1}{t - \psi_1} + \frac{\phi_2}{t - \psi_2} + \ldots + \frac{\phi_\nu}{t - \psi_\nu} \right) \end{aligned}$$

so that $\lambda(t)$ is an integral function of degree $\nu - 1$ in t. Since $\lambda(t)$ remains unaltered under every substitution of s, its coefficients are rational symmetric functions of $x_1, x_2, \ldots, x_n$ and hence are rational functions of the elementary symmetric functions.

Now we're assuming here that $x_1, x_2, \ldots, x_n$ denote indeterminant quantities since $\psi_1, \psi_2, \ldots, \psi_n u$ are algebraically distinct, so that $g'(\psi)$ is not idendically zero. In the case where special values are assigned to $x_1, x_2, \ldots, x_n$ such that two or more of the functions $\psi_1, \psi_2, \ldots, \psi_n u$ become numerically equaly, then $g'(\psi) = 0$, and ϕ is not a rational function of $\psi, c_1, ..., c_n$. Assuming that we're working with indetermiant $x_1, x_2, \ldots, x_n$, and taking $\psi_1 = \psi$ for t, we get

$$\lambda(\psi_1) = [(\psi_1 - \psi_2)(\psi_1 - \psi_3) \ldots (psi_1 - \psi_n u)] \, \phi_1$$

and in general

$$\phi = \frac{\lambda(\psi)}{g'(\psi)}.$$

Theorem 13.15 *Every two valued function of n variables is of the form $S_1 \pm S_2 \sqrt{\triangle}$, where S_1 and S_2 are integral symmetric functions, and $\triangle$ is the discriminant.*

Proof A two-valued function must belong the the group of order $\frac{1}{2}N$. The only group of this order is the alternating group to which the function $\sqrt{\triangle}$ belongs. The theorem therefore follows as an immediate consequence of the fundamental theorem Thm (13.13).

An Second Proof
Let two values of the function be denoted by ϕ_1 and ϕ_2 and let G_1 and G_2 be the corresponding groups, each of order $\frac{1}{2}N$. In the first place, these two groups must be identical; for it any substitution S of G_1 were to change ϕ_2 to its second value ϕ_1, then S^{-1} would change ϕ_1 into ϕ_2; but this is impossible, since S^{-1} and S each belong to G_1. Every substitution therefore of G_1 must belong to G_2 and vice-versa.

To show now that these groups coincided with the alternating group, consider the function $\psi = \phi_1 - \phi_2$. Any substitution which belongs to the common group leaves this unaltered; any other will change phi_1 to ϕ_2 and ϕ_2 to ϕ_1, and will therefore change the sign of ψ; some transposition, $(x_\alpha x_\beta)$ for example, will have this affect, for no group can include all the transpositions without coinciding with the symmetric group. It is easily inferred that $\psi_1 - \psi_2$ is divisible by $x_\alpha - x_\beta$, and hence by the product of all the differences, since ψ^2 is symmetric.

The quotient of ψ by $\sqrt{\triangle}$ is symmetric. To prove this, let $(\sqrt{triangle})^m$ be the highest power of $\sqrt{\triangle}$ which occurs in ψ. The quotient of ψ by $(\sqrt{\triangle}^m$ is symmetric, since, if not, it would be an alternating function, and would again contain $\sqrt{\triangle}$ as a factor, which is contrary to assumption. It follows immediately that m is an odd number and that the quotient of ψ by $\sqrt{\triangle}$ is symmetric. Writing therefore $\psi_1 - \psi_2 = A\sqrt{\triangle}$ and $\phi_1 + \phi_2 = B$, where A and B are both symmetric, we have

$$\phi_1 = S_1 + S_2\sqrt{\triangle} \text{ and } \phi_2 = S_1 - S_2\sqrt{\triangle},$$

where S_1 and S_2 are both symmetric functions of the variables $x_1, x_2, \ldots, x_n$. The groups G_1 and G_2 obviously coincide with the group of $\sqrt{\triangle}$ – the Alternatng Group.

Theorem 13.16 *The alternating functions are the only unsymmetrical functions of n variables of which a power can be symmetric.*

Proof It is sufficient to prove this theorem for prime powers; for if there exists a function $F(x_1, x_2, \ldots, x_n)$ such that F^{pq} is symmetric and p is a prime, then there is also a function $\phi = F^q$ such that ϕ^p is symmetric.

Therefore let

$$\phi^p = S, \text{ a symmetric function.}$$

Since the group of ϕ which is unsymmetric, cannot contain all the transpositions, let $\sigma = (x_\alpha x_beta)$ be a transposition which converts ϕ into ϕ_j; we have

$$\phi_j^p = \phi^p = S.$$

and therefore $\phi_j = \omega\phi$, where ω is a pth root of unity. Hence

$$\sigma\phi = \phi_j = \omega\phi,$$

and operating again with σ

$$\sigma^2\phi = \omega\sigma\phi = \omega^2\phi,$$

but $\sigma^2 = I$ and therefore $\omega^2 = 1$, and consequently $p = 2$.

Since ϕ^s is symmetric, ϕ is an alternating function. ∎

Theorem 13.17 *For any number, n, of independent elements there is no multiple-valied function of which a power is two-valiued when $n > 4$; and when $n = 3$, or $n = 4$, if there is any such power it is third power.*

Concentrating on prime numbers, and supposing that ϕ is a multiple-valued function whose p power is two-valued we have

$$\phi^p = S_1 + S_2\sqrt{\triangle}. \tag{41}$$

The group of ϕ cannot contain all the circular substitutions of the third order, for if it did, then this group would coincide with the alternating group, and ϕ would be two-valued. Let $\sigma = (x_\alpha, x_\beta, x_\gamma)$ be such a substitutions not contained in the gorup of ϕ and suppose $\sigma\phi = \phi_j$. From Eq. (41), since $S_1 + S_2\sqrt{\triangle}$ is unaltered by σ, we have

$$\psi^p = \phi_j^p,$$

hence $\phi = \omega\phi$, where ω is a p^{th} root of unity. Operating again twise in succession with σ we obtain

$$\begin{array}{rcl} \sigma\phi & = & \omega\phi, \\ \sigma^2\phi & = & \omega\sigma\phi = \omega^2\phi, \\ \sigma^3\phi & = & \omega^2\sigma\phi = \omega^3\phi, \end{array}$$

whence, since $\sigma^3 = 1$ (three cycle), we have $\omega^3 = 1$, and therefore $p = 3$.
Again, when the number of elements is greater than 4, there are circular substitutions of the fifth order, and those cannot all be contained in the group of ϕ (or else it will be two-valued and belong to the Alternating Group). Let τ be one of those not contained in this group and $\tau\phi = \phi_j$. We have, as before, from Eq. (41), by applying this substitution (which does not affect the right-hand side),

$$\psi^p = \phi_j^p = S_1 + S_2\sqrt{\triangle}.$$

Hence proceeding as before, we have $\tau\phi = \omega\phi$ and oeprating again on this and the succeeding equations with τ we easily find $\tau^5\phi = \omega^5\phi$; whence $\omega^5 = 1$, since $\tau^5 = 1$, and it is proved that $p = 5$. Now this results being incompatible with the previous value of p obtained, we infer that when the number of elements is greater than 4, it is impossile to find any multiple-valiued function ϕ, a prime power of which will be two-valued.

There are actually, when n is not greater than 4, multiple-valiued functions, a third power of which is two-vaued.

14 Some Results in the Theory of Numbers

Definition If a and b are two integers such that $a - b$ is divisible by n, we say that a is congruent to b modulo n, and write $a \equiv b \mod n$, or merely $a \equiv b(n)$.

Theorem 14.1 *Lagrange's theorem.*
The order of any subgroup of a finite group is a factor of the order of the group.
The order of a finite group G is a multiple of the order of every one of its subgroups.

Suppose H is a subgroup of G, where the order of G is n and that of H is m.
The whole group G may be decomposed into cosets gH relative to H. The number of elements in each coset must be

m, since the coset gH is formed by taking all the elements of H and pre-multiplying them by g, and all the elements so formed are different by the Cancellation Law. But the cosets are completely disjoint. Hence if there are r cosets we must have that $n = mr$, and so m is a factor of n.

Each element a of G generates a cyclic subgroup, whose order is simple the order of a. Therefore we have:

Corollary 14.2 *(Corollary 1): Every element of a finite group G has as order a divisor of the order of G.*

Corollary 14.3 *(Corollary 2): Every group G of prime order p is cyclic.*

Proof For the cyclic subgroup A generated by any element $a \neq e$ in such a group has an order $n > 1$ dividing p. But this implies $n = p$, and so $G = A$ is cyclic. ∎

Corollary 14.4 *(Corollary 3): The only abstract groups of order four are the cyclic group of that order and the four-group (Vierergruppe).*

Proof If a group G has order 4, contains and element of order 4, it is cyclic. Otherwise, by Corollary 1, all elements of G except e must have order 2. Call them a, b, c. By the cancellation law, ab cannot be either $ac = a, eb = b$, of $aa = e$; hence $ab = e$. By symmetry, $ac = ca = b, bc = cb = a, ba = c$. But these, together with $a^2 = b^2 = c^2 = e$, and $ex = xe = x$ for all x, give the multiplication table of the four-group. ∎

Theorem 14.5 *(Fermat's Little Theorem): If p is prime, $a^p \equiv a(p)$, and if a is not a mulltiple of p, then $a^{p-1} \equiv 1(p)$*

Proof (1) The numbers $a, 2a, 3a, ..., (p-1)a$ are all different modulo p if a is not a multiple of p, and so they must be $1, 2, ..., (p-1)$ modulo p in some order. Thus

$$a.2a.3a...(p-1)a \equiv 1.2.3...(p-1).$$

Hence $a^{(p-1)} \equiv 1(p)$. Thus $a^p \equiv a(p)$ if a is not a multiple of p, and this latter result is true also if a is a multiple of p, since then both sides are 0. Hence it is true for all a. ∎

Proof (2) The multiplication group mod p (excluding zero) has $(p-1)$ elements. The order of any element a of this group is then a divisor of $(p-1)$, by Corollary 1, so that $a^{(p-1)} \equiv 1(p)$ whenever $a \neq 0(p)$. If we multiply by a or both sides, we obtain the desired congruence, except for the case $a \equiv 0(p)$, for which the conclusion is trivially true. ∎

Proof (3) For a fixed prime p, let $P(n)$ be the proposition that $n^p \equiv n(p)$. Then $P(0)$ and $P(1)$ are obvious. In the binomial expansion for $(n+1)^p$, every coefficient except the first and the last is divisible by p, hence $(n+1)^p \equiv n^p+1(p)$, whence $P(n)$ implies $(n+1)^p \equiv n+1(p)$, which is the proposition $P(n+1)$. ∎

Theorem 14.6 *(Wilson's Theorem): Let $p > 2$ be a prime number. The field $\mathbb{Z}_p$ is the splitting field of $x^p - x$.*

Proof Since $\mathbb{Z}_p$ is the splitting field of $x^p - x$, we have

$$x^p - x = x(x-1)(x-2)\dots(x-(p-1)).$$

Thus

$$x^{(p-1)} - 1 = (x-1)(x-2)\dots(x-(p-1)),$$

and substituting $x = 0$ gives

$$-1 = (-1)(-2)\dots(-(p-1)).$$

There are an even number of factos on the right hand side, so the minus signs cancel out, leaving

$$(p-1)! \equiv -1(\mod p),$$

which is Wilson's Theorem.

15 Chinese Remainder Theorem

Consider a sequence of congruence equations:

$$\begin{aligned} x &\equiv a_1 \pmod{n_1} \\ &\vdots \\ x &\equiv a_k \pmod{n_k}, \end{aligned}$$

where the n_i are coprime in pairs. Let $N = n_1 n_2 \dots n_k$ and define $N_i = N/n_i$ be the product of all moduli but one. As the n_i are pairwise coprime, N_i and n_i are coprime. Therefore, by the Bezout identity, there exist integers M_i and m_i such that

$$M_i N_i + m_i n_i = 1.$$

Therefore a solution to the system of congruences is

$$x = \sum_{i=1}^{k} a_i M_i N_i.$$

Definition We restate the result as: if the n_i are pairwise coprime, the map

$$x \pmod N \to (x \pmod{n_1}, \dots, x \pmod{n_k})$$

defines a ring homomorphism

$$\mathbb{Z}/N\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \dots \times \mathbb{Z}/n_k\mathbb{Z}$$

between the ring of integers modulo N and the direct product of rings of integers modulo the n_i. This means that for doing a sequence of arithmetic operations in $\mathbb{Z}/N\mathbb{Z}$, one may do the same computation independently in each $\mathbb{Z}/n_i\mathbb{Z}$ and then get the result by applying the isomorphism (from the right to the left).

$$R/(\cap A_i) \cong R/(A_1) \times \dots \times R/(A_k)$$

We have a surjective map and know the kernel is the intersection of all the A_i's. If you take a map and quotient by the kernel, then that is isomorphic to the image – which is the whole set (everything) because the map is surjective. Therefore the LHS is isomorphic to the RHS.

16 Commutators

Definition Let G be a group. An element $g \in G$ is called a commutator if $g = aba^{-1}b^{-1}$ for elements $a, b \in G$. The smallest subgroup that contains all commutators of G is called the commutator subgroup or derived subgroup of G, and is denoted by G'.

Proposition 16.1 *Let G be a group with commutator subgroup G'.*

(a) The subgroup G' is normal in G, and the factor group G/G' is Abelian.

(b) If N is any normal subgroup of G, then the factor group G/N is Abelian, if and only if $G' \in N$.

Proof (a) Let $x \in G'$ and let $g \in G$. Then $gxg^{-1}x^{-1} \in G'$, and since $x \in G'$, we must have $gxg^{-1} = gxg^{-1}x^{-1}x \in G'$.

The factor group G/G' must be Abelian since for any cosets aG', bG', because G' is normal, we have $xG' = G'x$ for all $x \in G$.

$$aG'bG'a^{-1}G'b^{-1}G' = abG'G'a^{-1}b^{-1}G'G' = aba^{-1}b^{-1}G' = G'.$$

Proof (b) Let N be a normal subgroup of G. If $N \supseteq G'$, then G/N is a homomorphic image of G/G' and must be Abelian. Conversely, suppose that G/N is Abelian. Then $aNbN = bNaN$ for all $a, b \in G$, or simply $aba^{-1}b^{-1}N = N$, showing that every commutator of G belongs to N. This implies $G' \subseteq N$. ∎

Theorem 16.2 *The Symmetric Group G on n letters is not solvable unless $n \leq 4$.*

Proof Let $G = S_0 \supseteq S_1 \supseteq S_2 \supseteq \ldots \supseteq S_0$ be any chain of subgroups, each normal in the preceeding with cyclic quotient-group S_{k-1}/S_k; we shall prove by induction on s that S_s must contain every 3-cycle (ijk). This will imply that $S_s > I$, and so, that G cannot be solvable. Since $S_0 = G$ contains every 3-cycle, it is sufficient to show that if S_{k-1} contains every 3-cycle, then so does S_k. First note that if the permutations ϕ and ψ are both in S_{k-1}, then their so-called commutator $\gamma = \phi^{-1}\psi^{-1}\phi\psi$ is in S_s. To see this, consider their images ϕ' and ψ' in S_{k-1}/S_k. The latter, being cyclic, is commutative; hence

$$\gamma' = \phi'^{-1}\psi'^{-1}\phi'\psi' = \phi'^{-1}\psi'^{-1}\psi'\phi' = I \text{ in } S_{k-1}/S_k,$$

which implies that $\gamma \in S_k$. But in the special case when $\phi = (ijk)$ and $\psi = (jkm)$, where i, j, k are given and l, m are any two other letters (such letters exist unless $n \leq 4$), we have

$$\gamma = (jli)(mkj)(ilk)(jkm) = (ijk) \in S_k, \text{ for all } i, j, k.$$

This proves that S_k contains every 3-cycle, as desired.

The quotient chain doesn't terminate on the Identity, therefore the group G is not solvable unless $n \leq 4$. The 3-cycles generate the Alternating group which is simple and has no normal subgroups.

17 Fields

Theorem 17.1 *(Fields have no non-trivial Ideals): If a field F has an ideal I, then $I = (0)$ or $I = (1) = F$.*

Proof Let $a \in I$ and $a \neq 0$. Then $aa^{-1} = 1 \in I$ and so $f.1 = f \in I$ for any $f \in F$ Therefore $F \subset I$. But $I \subset F$ by definition of Ideals. Therefore $I = F$. The only other possibility is $a = 0$ in which case we have the trivial Ideal $I = (0) = 0$.

If F is any field, then the smallest subfield of F that contains the identify element 1 is called the *prime subfield* of F. If F is a finite field, then its prime subfield is isomorphic to $\mathbb{Z}_p$, where $p = \text{char}(F)$ for some prime p. When we want to emphasize $\mathbb{Z}/p\mathbb{Z}$ with its field structure, we denote it by $\mathbb{F}_p$. The prime subfield of $\mathbb{Z}_p$ which is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ is $\mathbb{F}_p$. The only prime fields are the fields of residues modulo p and the field of rationals.

The prime subfield of a field F is the intersection of all subfields of F. The intersection of any two subfields is a field and similarly for all the other subfields. Also the intersection of all subfields has no proper subfields, since any such would be subfields of R and hence must contain the intersection which is impossible. Hence the intersection is a prime subfield. F cannot contain two distinct prime subfields, since their intersection would also be a subfield and hence must be identical with both.

18 Finite Fields

Theorem 18.1 *(Characteristic of a Field): In a field F, all non-zero elements have the same additive order. This is called the* characteristic *of F and is said to be zero if all non-zero elements have infinite order.*

Proof Suppose that any given non-zero element y has finite order m. Then if x is any other non-zero element, $(mx)y = m(xy) = x(my) = 0$. But $y \neq 0$ and so $mx = 0$. This is a step that could not be performed in the case of rings, which may have zero divisors. Also, suppose $rx = 0$ for some $r < m$. Then $x(ry) = (rx)y = 0$ as before and since $x \neq 0$, $ry = 0$, which is impossible since $r < m$ and m is the order of y. Hence x also has order m.

Theorem 18.2 *If F has a finite characteristic, this must be a prime.*

Proof Suppose F has as characteristic a composite number hk. If a is any non-zero element of F, a^2 is non-zero and so has order hk, so that $hka^2 = 0$, i.e., $(ha)(ka) = 0$ and so either ha or ka is 0, which is impossible, since both h and k are less than hk and a must have order hk.

Theorem 18.3 *If F has characteristic p, it contains a subfield isomorphic to the field of residues modulo p.*

Proof Consider the unity 1 and denote 1+1 by 2, 1+1+1 by 3, etc. Then 1 has order p, and so $p.1 = 0$ (i.e., the sum of the p 1's is the zero element). Now consider the subset of elements $0, 1, 2, \dots, (p-1)$. The sum and difference of any two of these is in the set as are 0 and 1. Furthermore the product of any two is in the set, since the product is the ordinary product of residues modulo p, by the Distributive Law. For example,

$$2.3 = (1+1)(1+1+1) = 1+1+1+1+1+1 = 6.$$

Since sum and product behave as they do for residues modulo p the subset is isomorphic to the set of residues modulo p under residue sum and product and so, since p is prime, inverses exist in the subset and the subset is isomorphic to the field of residues modulo p.

Theorem 18.4 *Let F be a field of characteristic 0, it contains a subfield isomorphic to the field of rational numbers.*

Proof The unity 1 has infinite order and so the elements $0, 1, 2 \dots$ are all distinct. Consider the subset of all elements m/n where m and n are co-prime, n is non-zero and m/n means $m.n^{-1}$. It is easily seen that this subset contains the sum, difference, product and quotient of any two of its elements (except dividing by 0, of course) and also contains 0 and 1, and so is a subfield, and is trivially isomorphic to the field of rationals.

Definition A finite field is a field which has a finite number of elements.

Theorem 18.5 *Any finite field has a characteristic p, for p a finite prime (i.e., it cannot have characteristic 0).*

For by Theorem 18.4 a field with characteristic 0 contains an infinite subfield and so must itself be infinite.

Theorem 18.6 *Let F be a finite field of characteristic p. The F has p^n elements, for some positive integer n.*

Proof Recall that if F has characteristic p, then the ring homomorphism $\phi : \mathbb{Z} \to F$ defined by $\phi(n) = n.1$ for all $n \in \mathbb{Z}$ has kernel $p\mathbb{Z}$, and thus the image of ϕ is a subfield K of F isomorphic to $\mathbb{Z}_p$. Since F is finite, it must certainly have finite dimension as a vector space over K, say $[F : K] = n$. If $v_1, v_2, \ldots, v_n$ is a basis for F over K, then each element of F has the form $a_1v_1 + a_2v_2 + \ldots + a_nv_n$ for elements $a_1, a_2, \ldots, a_n \in K$. Thus to define an elements of F there are n coefficients a_i, and for each coefficient there are p choices, since K has only p elements. Therefore the total number of ways in which an element in F can be defined is p^n. ∎

Theorem 18.7 *Let F be a finite field with p^n elements. Then F is the splitting field of the polynomial $x^{p^n} - x$ over the prime subfield of F.*

Proof Let F be a finite field of characteristic p. Then as in Theorem 18.6 the field F is an extension of degree n of its prime subfield K, which is isomorphic to $\mathbb{Z}_p$. Since F has p^n elements, the order of the multiplicative group $F^\times$ of non-zero elements of F is $p^n - 1$. Therefor $x^{p^n-1} \equiv 1$ for all $0 \neq x \in F$, and so $x^{p^n} = x$ for all $x \in F$. The polynomial $f(x) = x^{p^n} - x$ has at most p^n roots, and so its roots must be precisely the elements of F. Thus F is the splitting field of $f(x)$ over K. ∎

19 Descartes Rule of Signs

Theorem 19.1 *(Descartes Rule of Signs): The number of variatons of sign in the subsequent terms of a polynomial, s, minus the number of positive roots, p, is a non-negative even integer.*

$$s - p = 2r, \quad r \geq 0$$

a) $s - p$ is an even integer.

Let $f(x)$ be a polynomial with a positive leading coefficient .

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n, \quad \text{and} \quad a_0 > 0.$$

Then since the sequence of signs starts and ends with a $+$ we must have an even number of variations $+-$ and $-+$ of sign in the sequence

$$+ \ldots + .$$

to get back eventually to find the final $+$ sign so s is even.

Similarly p is even because the function must start at $y = a_0$ and can only cross the x-axis an even number of times to eventually again be in the $y > 0$ region. If there are multiple roots of the form $(x - \alpha)^k = 0$, then these are counted with their respective multiplicity. If k is even, then the function does not cross the x-axis at the multiple root. If k is odd, then the function crosses the x axis and goes into the $y < 0$ region. In either case, p is even. Hence if $a_0 > 0$, we have that $s - p$ is even.

Similar arguments hold for the case when $a_0 < 0$. In that case both s and p are odd numbers and therefore $s - p$ is still even.

b) $s - p$ is non-negative.

Use induction on the degree of the polynomial.

$n = 1$. $x + a_n = 0$. Zero variations if a_n is positive and zero positive roots so $s - p = 0$ is even. If $a_n < 0$ then there is one variation in sign and one positive root, $s = 1$ and $p = 1$ and $s - p = 0$.

Induct on $n = k$.

$$f(x) = a_0 x^k + a_1 x^{k-1} + a_2 x^{k-2} + \ldots + a_{k-1} x + a_k.$$

Take the derivative with respect to x,

$$f'(x) = k a_0 x^{k-1} + (k-1) a_1 x^{k-2} + (k-2) a_2 x^{k-3} + \ldots + a_{k-1}.$$

Now, since the k terms that multiply the various x powers are all positive then

$$s = \text{sign variation}[a_0, a_1, a_2, \ldots, a_{k-1}, a_k]$$

and

$$s' = \text{sign variation}[a_0, a_1, a_2, \ldots, a_{k-1}]$$

Therefore

$$s \geq s'.$$

Since the roots p' are interspersed between the roots of $f(x)$ by Rolle's theorem, the number of positive roots can be $p - 1$ which occur between the roots of $f(x)$ plus possibly one more between $x = 0$ and the first root of α_1 of $f(x)$, so:

$$p' \geq p - 1.$$

Therefore we have

$$s \geq s' \geq p' \geq p - 1$$

or

$$s - p \geq -1.$$

Part a) shows that $s - p$ must be an even number in all cases, so $s - p \geq -1 \implies s - p \geq 0$. Hence

$$s - p = 2r, \quad r \geq 0.$$

20 Resultant of Two Polynomials

Following the demonstration in [3], consider two polynomials

$$f(x) = \sum_{i=0}^{n} f_i \, x^{n-i} = f_0 x^n + f_1 x^{n-1} + f_2 x^{n-2} + \ldots + f_{n-1} x + f_n$$

$$g(x) = \sum_{j=0}^{m} g_j \, x^{m-j} = g_0 x^m + g_1 x^{m-1} + g_2 x^{m-2} + \ldots + g_{m-1} x + g_m$$

Let use assume that $f(x)$ and $g(x)$ have a common factor $(x-\alpha)$. Then

$$\begin{aligned} f(x) &= (x-\alpha)f_1(x) \quad \text{and} \\ g(x) &= (x-\alpha)g_1(x). \end{aligned}$$

Then

$$f(x)g_1(x) = g(x)f_1(x) = \frac{f(x)g(x)}{x-\alpha}.$$

Assuming that

$$\begin{aligned} f_1(x) &= -\sum_{i=0}^{n-1} z_i\, x^{n-1-i} = -(z_0x^{n-1} + z_1x^{n-2} + z_2x^{n-3} + \ldots + z_{n-2}x + z_{n-1}) \\ g_1(x) &= \sum_{j=0}^{m-1} y_j\, x^{m-1-i} = y_0x^{m-1} + y_1x^{m-2} + y_2x^{m-3} + \ldots + y_{m-2}x + y_{m-1} \end{aligned}$$

By equating like powers of x, we can write out a linear system corresponding to the equation

$$f(x)g_1(x) - g(x)f_1(x) = 0$$

we find

$$\begin{aligned} -g(x)f_1(x) = \; & g_0z_0x^{n+m-1} + g_0z_1x^{n+m-2} + g_0z_2x^{n+m-3} + \ldots + g_0z_{n-2}x^{m+1} + g_0z_{n-1}x^m + \\ & g_1z_0x^{n+m-2} + g_1z_1x^{n+m-3} + \ldots + g_1z_{n-3}x^{m+1} + g_1z_{n-2}x^m + g_1z_{n-1}x^{m-1} + \\ & g_2z_0x^{n+m-3} + g_2z_1x^{n+m-4} + \ldots + g_2z_{n-3}x^m + g_2z_{n-2}x^{m-1} + g_2z_{n-1}x^{m-2} + \\ & \vdots \;\; + \\ & \ldots + (g_{m-2}z_{n-1} + g_{m-1}z_{n-2} + g_mz_{n-3})x^2 + (g_{m-1}z_{n-1} + g_mz_{n-2})x + g_mz_{n-1} \end{aligned}$$

$$\begin{aligned} g(x)g_1(x) = \; & f_0y_0x^{n+m-1} + f_0y_1x^{n+m-2} + f_0y_2x^{n+m-3} + \ldots + f_0y_{m-2}x^{n+1} + f_0y_{n-1}x^n + \\ & f_1y_0x^{n+m-2} + f_1y_1x^{n+m-3} + \ldots + f_1y_{n-3}x^{n+1} + f_1y_{m-2}x^n + f_1y_{n-1}x^{n-1} + \\ & f_2y_0x^{n+m-3} + f_2y_1x^{n+m-4} + \ldots + f_2y_{n-3}x^n + f_2y_{m-2}x^{n-1} + f_2y_{n-1}x^{n-2} + \\ & \vdots \;\; + \\ & \ldots + (f_{n-2}y_{m-1} + f_{n-1}y_{m-2} + f_ny_{m-3})x^2 + (f_{n-1}y_{m-1} + f_ny_{m-2})x + f_ny_{m-1} \end{aligned}$$

In order to satisfy $f(x)g_1(x) - g(x)f_1(x) = 0$, all the terms multiplying like powers of x each must vanish. This can be arranged as a linear system in the vector $[z_0, z_1, ..., z_{n-1}, y_0, y_2, ...y_{m-1}]$. The coefficient matrix of this linear system is equal to the transpose of the Sylvester Resultant matrix which is defined as

$$R_{m,n}(g,f) = \underbrace{\begin{bmatrix} g_0 & \cdots & g_m & & \\ & \ddots & \cdots & \ddots & \\ & & g_0 & \cdots & g_m \\ f_0 & \cdots & f_n & & \\ & \ddots & \cdots & \ddots & \\ & & f_0 & \cdots & f_n \end{bmatrix}}_{m+n} \begin{matrix} \left.\vphantom{\begin{matrix} a\\ a\\ a\end{matrix}}\right\} n \text{ lines} \\ \left.\vphantom{\begin{matrix} a\\ a\\ a\end{matrix}}\right\} m \text{ lines} \end{matrix}$$

The corresponding linear system that sets all the coefficients of the same powers of x to 0 in $f(x)g_1(x)-g(x)f_1(x)=0$ is

$$\begin{bmatrix} g_0 & \cdots & g_m & & \\ & \ddots & \cdots & \ddots & \\ & & g_0 & \cdots & g_m \\ f_0 & \cdots & f_n & & \\ & \ddots & \cdots & \ddots & \\ & & f_0 & \cdots & f_n \end{bmatrix}^T \begin{bmatrix} z_0 \\ \vdots \\ z_{n-1} \\ y_0 \\ \vdots \\ y_{m-1} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The criterion for a non-trivial solution to the above linear system is

$$\det \begin{bmatrix} g_0 & \cdots & g_m & & \\ & \ddots & \cdots & \ddots & \\ & & g_0 & \cdots & g_m \\ f_0 & \cdots & f_n & & \\ & \ddots & \cdots & \ddots & \\ & & f_0 & \cdots & f_n \end{bmatrix} = 0$$

The rank of the matrix $R_{m,n}(g,f)$ also gives the multiplicity of the common root of $f(x)$ and $g(x)$. The leading term of this determinant is $g_0^n f_n^m$. The terms of in the expansion an $(n+m)\times(n+m)$ matrix M of the form $\sum_\sigma \epsilon_{r_1,r_2,r_3,\ldots r_{n+m}} m_{1r_1} m_{2r_2} m_{3r_3} \ldots m_{(n+m)r_{n+m}}$ where the sum is over all permutations σ of the numbers $[1,2,\ldots,n+m]$ and $sgn(\sigma) = \epsilon_{r_1,r_2,r_3,\ldots r_{n+m}}$ is the sign of the permutation. In the case of the Sylvester Resultant the terms take the form

$$sgn(\sigma) g_0^{\nu_0} g_1^{\nu_1} \ldots g_m^{\nu_m} f_0^{\mu_0} f_1^{\mu_1} \ldots f_n^{\mu_n},$$

where

$$n = \nu_0 + \nu_1 + \nu_2 + \ldots + \nu_m$$
$$m = \mu_0 + \mu_1 + \mu_2 + \ldots + \mu_n$$

Factoring out $g_0^n f_0^m$ the generic term in the expansion of the determinant becomes

$$sgn(\sigma) g_0^n f_0^m (\frac{g_1}{g_0})^{\nu_1} (\frac{g_2}{g_0})^{\nu_2} \ldots (\frac{g_m}{g_0})^{\nu_m} (\frac{f_1}{f_0})^{\mu_1} (\frac{f_2}{f_0})^{\mu_2} \ldots (\frac{f_n}{f_0})^{\mu_n}$$

Hence the Sylvester Resultant has the form

$$R_{m,n}(g,f) = g_0^n f_0^m \phi(\alpha,\beta)$$

where ϕ is an integral function of the roots α_i and β_i. As the determinant vanishes for any common roots $\alpha_i = \beta_k$, the determinant is divisible by $(\alpha_i - \beta_k)$ and the determinant is divisible by

$$g_0^n f_0^m \prod_{i,k} (\alpha_i - \beta_k)$$

Theorem 20.1 *$R_{m,n}(g, f)$ is a homogenous polynomial with integer coefficients g_i, f_j.*

(i) $R_{m,n}(g, f)$ is homogenous of degree n in $g_m, ..., g_0$ and degree m in $f_n, ..., f_0$.

(ii)If g_i and f_i are regarded as having degree i. then $R_{m,n}(g, f)$ is homogenous of degree mn.

Proof It is obvious that $R_{m,n}(g, f)$ is a homogenous polynomial with integer coefficients in the coefficients g_i, f_j, of total degree $m+n$. Moreover, to replace g_i by tg_i and f_j by uf_j in the Sylvester matrix means that we multiply each of the first m rows by t and each of the last n rows by u, and thus the determinant $R_{m,n}(g, f)$ by $t^m u^n$, which shows part (i). It is here best to treat g_i, f_j, t and u as different inderminants a do the calculations in $F(g_0, ..., g_m, f_0, ..., f_n, t, u)$. Similarly, (ii) follows because to replace each g_i by $t^i g_i$ and each f_j by $t^j f_j$ in $R_{m,n}(g, f)$ yields the same result as multiplying the ith row by t^{m+i} for $i = 1, 2, ..., m$ and by t^i for $i = m+1, ..., m+n$, and the jth column by t^{-j}. The overall effect it to multiply the determiant $R_{m,n}(g, f)$ by t^K where $K = mn + \sum_{i=1}^{m+n} i - \sum_{j=1}^{m+n} j = mn$. ∎

21 Linear Spaces

A linear space $\mathbb{L}_p$ of dimension p is the totality of points of $\mathbb{R}_n$ into which a fixed point P is carried by the vectors of a p-dimensional vector space L of $\mathbb{R}_n$. Suppose there are given $p+1$ points

$$P_k = (x_k^1, x_k^2, ..., x_k^n), \; k = 0, 1, ..., p$$

which do not lie an any linear space of dimension $< p$. Let us consider the vectors $\mathfrak{a}_k = \overrightarrow{P_0 P_k}, k = 1, 2, ..., p$. These p vectors are linearly independent. For if they were linearly dependent, then the vector space spanned by them would have dimension $< p$. By applying these vectors to this vector space to P_0, we obtain a linear space which on the one hand has dimension $< p$, and on the other contains the points $P_0, P_1, ..., P_p$, which contradicts our assumption.

All points of a linear space are equivalent to one another because any point of it can be used to obtain the entire linear space.

22 Linear Equations

Let a system of m linear equations in n unknowns $x^1, x^2, ..., x^n$ be given in the following form:

$$\begin{array}{cccc} a_{11}x^1 & + a_{12}x^2 & + \ldots & +a_{1n}x^n = b_1 \\ a_{21}x^1 & + a_{22}x^2 & + \ldots & +a_{2n}x^n = b_2 \\ \cdots & \cdots & \cdots & \cdots\cdots \\ a_{i1}x^1 & + a_{i2}x^2 & + \ldots & +a_{in}x^n = b_i \\ \cdots & \cdots & \cdots & \cdots\cdots \\ a_{m1}x^1 & + a_{m2}x^2 & + \ldots & +a_{mn}x^n = b_m \end{array} \tag{42}$$

which can be written in the following form based on the rules of matrix multiplication

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x^1 \\ x^2 \\ . \\ x^i \\ . \\ x^n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ . \\ b_i \\ . \\ b_m \end{bmatrix} \tag{43}$$

To every column of the matrix of coefficients we can associate a vector

$$\mathfrak{a}_k = \{a_{1k}, a_{2k}, ..., a_{mk}\},\ \ k = 1, 2, ..., n.$$

The original system of equations can be expressed as a single vector equation

$$\mathfrak{a}_1 x^2 + \mathfrak{a}_2 x^2 + ... + \mathfrak{a}_n x^n = \mathfrak{b}, \tag{44}$$

where $\mathfrak{b} = \{b_1, b_2, ..., b_m\}$.

Consider the vector space $\mathbb{L}$ spanned by the vectors $\mathfrak{a}_k, k = 1, ..., n$ in $\mathbb{R}_m$. If Eq.(44) is solvable for $x^1, x^2, ..., x^n$, then $\mathfrak{b}$ must be a linear combination of the $\mathfrak{a}_k$ and therefore belongs to $\mathbb{L}$. Conversely if $\mathfrak{b}$ belongs to $\mathbb{L}$ then Eq.(44) is solvable.

Consider the vector space $\mathbb{L}'$ spanned by the vectors $\mathfrak{a}_1, \mathfrak{a}_1, ..., \mathfrak{a}_n, \mathfrak{b}$. A vector $\mathfrak{c}$ in $\mathbb{L}'$ is of the form

$$\mathfrak{c} = \lambda_1 \mathfrak{a}_1 + \lambda_2 \mathfrak{a}_2 + ... + \lambda_n \mathfrak{a}_n + \lambda \mathfrak{b}.$$

If $\mathfrak{b}$ belongs to $\mathbb{L}$, then Eq. (44) is satisfied by certain numbers $x^1, x^2, ..., x^n$ which can be substituted for $\mathfrak{b}$ and so $\mathfrak{c}$ is a linear combination of the $\mathfrak{a}_k, k = 1, ..., n$ and therefore belongs to $\mathbb{L}$. In this case $\mathbb{L}' \subseteq \mathbb{L}$ and also $\mathbb{L} \subseteq \mathbb{L}'$ and therefore $\mathbb{L}'$ and $\mathbb{L}$ are identical.

Theorem 22.1 *The equation (44) is solvable for the x^i, if and only if, the maximal number of linearly independent vectors among $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n$ is the same as the maximal number of linearly independent vectors among $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_n, \mathfrak{b}$.*

Definition The **rank** of a matrix is equal to the maximal number of independent column vectors in the matrix.

Consider the following matrix

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{in} & b_i \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

obtained by appending the vector $\{b_1, b_2, ..., b_m\}$ to the final column. This matrix is called the **augmented** matrix of the system of equations (42).

Theorem 22.2 *Eqs. (42) are solvable for the x^i if and only if the rank of the coefficient matrix of the system (42) is equal to the rank of the augmented matrix*

The following system of equations is called a **homogenous** *in the* x^i

$$\begin{array}{cccc} a_{11}x^1 & + a_{12}x^2 & +\dots & +a_{1n}x^n = 0 \\ a_{21}x^1 & + a_{22}x^2 & +\dots & +a_{2n}x^n = 0 \\ \cdots & \cdots & \cdots & \cdots\cdots \\ a_{i1}x^1 & + a_{i2}x^2 & +\dots & +a_{in}x^n = 0 \\ \cdots & \cdots & \cdots & \cdots\cdots \\ a_{m1}x^1 & + a_{m2}x^2 & +\dots & +a_{mn}x^n = 0 \end{array} \tag{45}$$

Any set of numbers $x^1, x^2, ..., x^n$ which satisfy Eq. (45) are now taken to be the components of an n-dimensional vector $\mathfrak{r} = \{x^1, x^2, ..., x^n\}$. We call such a vector a *vector solution* of the system (45). The totality of vector solutions of the system of equations (45) forms a vector space.

Theorem 22.3 *If the coefficient matrix of the system of equations (45) has rank r, then the set of vector solutions of this system is an $(n-r)$-dimensional vector space.*

Theorem 22.4 *The system of equations (45) has a non-trivial solution, i.e. a solution $x^1, x^2, ..., x^n$ such that not all the x^i vanish, if and only if, the rank of the matrix of (45) is less than n.*

Theorem 22.5 *If the number of equations in the system of homogenous equations (45) is less than the number of unknowns, then the system must have non-trivial solutions.*

Theorem 22.6 *All the solutions of a non-homogenous system of equations (42) are of the form $z = \mathfrak{r} + \eta$, where $\mathfrak{r}$ is a fixed solution of the non-homogenous system (42) and η runs though all solutions of the corresponding homogenous system (45).*

Let s be the maximal number of linearly independent rows of the matrix row vectors. Let us assume that the rows of the homogenous system are ordered so that the first s rows are independent to start with. This involves no loss of generality since it does not affect that rank. The i-th equation of this system is, by definition of the scalar product of two vectors, equivalent to the vector equation

$$\mathfrak{a}_i \cdot \mathfrak{r} = 0,$$

where we set $\mathfrak{r} = \{x^1, x^2, ..., x^n\}$.
Every system of n numbers $x^1, x^2, ..., x^n$ satisfying the first s equation of (the re-ordered) (45), satisfies all of the equations of (45). Because the first s-rows are linearly independent, all the rows satisfy a relation of the form

$$\mathfrak{a}_k = \lambda_1^k \mathfrak{a}_1 + \lambda_2^k \mathfrak{a}_2 + ... + \lambda_s^k \mathfrak{a}_s, \text{ for } 1 \leq k \leq m.$$

If we apply the distributive law on the right, we obtain

$$\mathfrak{a}_k \cdot \mathfrak{r} = \lambda_1^k (\mathfrak{a}_1 \cdot \mathfrak{r}) + \lambda_2^k (\mathfrak{a}_2 \cdot \mathfrak{r}) + ... + \lambda_s^k (\mathfrak{a}_s \cdot \mathfrak{r}).$$

Since $\mathfrak{a}_i \cdot \mathfrak{r} = 0$ for $i = 1, 2, ..., s$ the right hand side of this last equation becomes 0, so that

$$\mathfrak{a}_k \cdot \mathfrak{r} = 0, \quad k = 1, 2, ..., m,$$

and in particular for $k = s+1, s+2, ..., m$ as was to be proved.

The vector space of all the solutions of the first s equations of (45) is thus identical with the space of solutions of the entire system (45). It dimension is this $n - r$ since r is the rank of the matrix of coefficients in (43). It thus follows that the rank of the matrix of the first s equations, i.e of the matrix which consists only of the first s rows of (45) is equal to $n-(n-r)=r$. But the column vectors of this matrix are vectors with s components. Thus the maximal number of linearly independent column vectors of this matrix is $\leq s$ since by previous theorem any $s+1$ vectors of s-dimensional vector space are linearly dependent.

Therefore the maximal number of linearly independent column vectors of a matrix is at most as large as the maximal number of linearly independent row vectors.

Now consider the transpose of the matrix of coefficients of (45). The maximal number of independent column vectors of this matrix is s, and the maximal number of row vectors is r. Applying the previous result we also obtain $s \leq r$. Hence we have $r \leq s$ and $s \leq r$ and therefore s = r. Thus we have proved

Theorem 22.7 *The maximal number of linearly independent column vectors of a matrix is equal to the maximal number of linearly independent row vectors of that matrix.*

Theorem 22.8 *With any given vector space L of $\mathbb{R}_n$, we can always associate a system of homogenous linear equations in n unknowns such that all the vectors of L, and no others, are vector solutions of this system.*

Let $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ be a basis of L, where we set $\mathfrak{a}_i = \{\mathfrak{a}_{i1}, \mathfrak{a}_{i2}, ..., \mathfrak{a}_{in}\}$. If in addition we set

$$\mathfrak{x} = \{x^1, x^2, ..., x^n\}$$

then the equations

$$\mathfrak{a}_i \cdot \mathfrak{x} = 0, \quad i = 1, 2, ..., p, \tag{46}$$

form a system of homogenous linear equations in the unknowns $x^1, x^2, ..., x^n$ whose matrix is

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \dots & \dots & \dots & \dots \\ a_{p1} & a_{p2} & \dots & a_{pn} \end{bmatrix}$$

Let p be the dimension of L.The rank of this system must be $n-p$. The totality of vectors which are orthogonal to L is a vector space L' of dimension $n-p$. It consists precisely of all the vector solutions of (46).

We now seek a those vectors which are othogonal to the vector space L' just found, We chose a basis $\mathfrak{b}_1, \mathfrak{b}_2, ..., \mathfrak{b}_{n-p}$ of L'.The $\mathfrak{b}_i$ as vectors of L' are orthogonal to L, and this in particular to $\mathfrak{a}_1, \mathfrak{a}_2, ...\mathfrak{a}_p$. Thus, for $i = 1, 2, ..., p$, we have

$$\mathfrak{b}_1 \cdot \mathfrak{a}_i = 0, \quad \mathfrak{b}_2 \cdot \mathfrak{a}_2 = 0, \quad ..., \quad \mathfrak{b}_{n-p} \cdot \mathfrak{a}_i = 0.$$

Now just as for L, the totality of all vectors orthogonal to L' consists of all the vector solutions of the equations

$$\mathfrak{b}_1 \cdot \mathfrak{x} = 0, \quad \mathfrak{b}_2 \cdot \mathfrak{x} = 0, \quad ..., \quad \mathfrak{b}_{n-p} \cdot \mathfrak{x} = 0. \tag{47}$$

Thus the $\mathfrak{a}_i, i = 1, 2, ..., p$ are orthogonal to L'. But the vectors which are orthogonal to L' form a vector space L'' of dimension $n-(n-p) = p$.Since the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are in L'' and are linearly independent, then form a basis of L''. Therefore L'' is identical with L. Thus the vectors of L are precisely the vector solutions of equations (47), i.e, of a certain system of $n-p$ linear homogenous equations.

Theorem 22.9 *The point solutions of a solvable system of equations of the type (42) form a linear space. This space has dimensions $n-rm$ where r is the rank of the coefficient matrix of the system. Conversely, every linear space may be represented as the totality of all point solutions of some suitable system of linear equations.*

23 Vector Spaces in $\mathbb{R}_n$

Definition Vectors are directed segments. The vector in $\mathbb{R}_n$ whose components are $a_1, a_2, ..., a_n$ is denoted by

$$\mathfrak{a} = \{a_1, a_2, ..., a_n\} \text{ (braces).}$$

The zero vector in $\mathbb{R}_n$ is denoted as

$$\mathfrak{o} = \{0, 0, ..., 0\},$$

with n components each 0.

Definition We call p vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ **linearly dependent** if there are p numbers $\lambda_1, \lambda_2, ..., \lambda_p$ not all 0, such that

$$\lambda_1\mathfrak{a}_1 + \lambda_2\mathfrak{a}_2 + ... + \lambda_p\mathfrak{a}_p = \mathfrak{o} \quad \text{(the zero vector).}$$

If no such p numbers exist, then the p vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are called **linearly independent**.

Theorem 23.1 *It a subset of the p vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ is linearly dependent, then the set of p vectors is itself linearly dependent.*

Suppose the first r vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_r$, $(r < p)$are linearly dependent. Then there are r numbers $\lambda_1, \lambda_2, ..., \lambda_r$ not all zero such that

$$\lambda_1\mathfrak{a}_1 + \lambda_2\mathfrak{a}_2 + ... + \lambda_r\mathfrak{a}_r = \mathfrak{o}.$$

Setting $\lambda_{r+1}, \lambda_{r+2}, ..., \lambda_p$ to zero we have

$$\lambda_1\mathfrak{a}_1 + \lambda_2\mathfrak{a}_2 + ... + \lambda_r\mathfrak{a}_r + \lambda_{r+1}\mathfrak{a}_{r+1} + ... + \lambda_p\mathfrak{a}_p = \mathfrak{o}.$$

where the numbers $\lambda_1, \lambda_2, ..., \lambda_p$ are not all zero. Therefore the p vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly dependent.

Theorem 23.2 *If the p vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly independent, then so is every subset of these p vectors.*

For otherwise, by Theorem 23.1, the p vectors would be linearly dependent.

Theorem 23.3 *If the p vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly dependent and if $p > 1$, then at least one of these vectors is a linear combination of the others.*

For, there are numbers λ_i, which do not all vanish such that

$$\lambda_1\mathfrak{a}_1 + \lambda_2\mathfrak{a}_2 + ... + \lambda_p\mathfrak{a}_p = \mathfrak{o}.$$

Suppose $\lambda_p \neq 0$. The we can solve for $\mathfrak{a}_p$,

$$\mathfrak{a}_p = -\frac{\lambda_1}{\lambda_p}\mathfrak{a}_1 - \frac{\lambda_2}{\lambda_p}\mathfrak{a}_2 - ... - \frac{\lambda_{p-1}}{\lambda_p}\mathfrak{a}_{p-1},$$

i.e. $\mathfrak{a}_p$ is a linear combination of the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_{p-1}$.

Theorem 23.4 *If one of the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ is a linear combination of the others, then vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly dependent.*

Suppose
$$\mathfrak{a}_1 = \lambda_2\mathfrak{a}_2 + \lambda_3\mathfrak{a}_3 + ... + \lambda_p\mathfrak{a}_p.$$
Then we have that
$$\mathfrak{a}_1 - \lambda_2\mathfrak{a}_2 - \lambda_3\mathfrak{a}_3 - ... - \lambda_p\mathfrak{a}_p = \mathfrak{o},$$
and since the coefficient of $\mathfrak{a}_1$ is not zero, the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly dependent.

Theorem 23.5 *If the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly independent, and if the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ...\mathfrak{a}_p, \mathfrak{b}$ are linearly dependent, then $\mathfrak{b}$ is a linear combination of $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$.*

We have a relation of the form
$$\lambda_1\mathfrak{a}_1 + \lambda_2\mathfrak{a}_2 + ... + \lambda_p\mathfrak{a}_p + \lambda_{p+1}\mathfrak{b} = \mathfrak{o},$$
where not all the λ_i vanish. Now λ_{p+1} cannot be 0 because the last term would drop out and then all the other λ_i would vanish because of the linear independence of the $\mathfrak{a}_i$ vectors. Therefore $\lambda_{p+1} \neq 0$ and so we have
$$\mathfrak{b} = -\frac{\lambda_1}{\lambda_{p+1}}\mathfrak{a}_1 - \frac{\lambda_2}{\lambda_{p+1}}\mathfrak{a}_2 - ... - \frac{\lambda_p}{\lambda_{p+1}}\mathfrak{a}_p.$$

Theorem 23.6 *If the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ are linearly independent, and if $\mathfrak{b}$ is a not a linear combination of $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$, then the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ...\mathfrak{a}_p, \mathfrak{b}$ are linearly independent.*

Theorem 23.6 follows from Theorem 23.5.
Let L be a *non-empty* set of vectors of $\mathbb{R}_n$ with the properties:

1. If $\mathfrak{a}$ is a vector of the set L, then $\lambda\mathfrak{a}$ belongs to L for every real λ.
2. If $\mathfrak{a}$ and $\mathfrak{b}$ are two (not necessarily distinct) vectors of the set L, then the vector $\mathfrak{a} + \mathfrak{b}$ also belongs to L.

Such a set of vectors, satisfying 1. and 2., is called a **vector space** in $\mathbb{R}_n$.

Theorem 23.7 Steinitz replacement theorem. *Let $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ be any p vectors, and let L be the vector space spanned by them. Let $\mathfrak{b}_1, \mathfrak{b}_2, ..., \mathfrak{b}_q$ be any q linearly independent vectors of L. Then we may replace a certain set of q vectors from among the vectors $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ by the vectors $\mathfrak{b}_1, \mathfrak{b}_2, ..., \mathfrak{b}_q$ so that the remaining vectors of $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ together with the vectors $\mathfrak{b}_1, \mathfrak{b}_2, ..., \mathfrak{b}_q$ span the entire vector space L.*

Proof Use mathematical induction.

Theorem 23.8 *Any set of more than p linear combinations of p given vectors is always linearly dependent.*

Theorem 23.9 *The maximal number of linearly independent vectors in $\mathbb{R}_n$ is n*

Definition Dimension and Basis
Let L be any vector space in $\mathbb{R}_n$. The maximal number of linearly independent vectors is $\leq n$. Let p be this maximal number of independent vectors in L. We call p the **dimension** of L. Thus,
$$0 \leq p \leq n.$$

We call any system of p linearly independent vectors of L, such as $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ a **basis** of L.

Theorem 23.10 *Every vector of* L *can be represented in exactly one way as a linear combination of the basis vectors* $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$.

Theorem 23.11 *The dimension of the vector space spanned by the vectors* $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$ *is equal to the maximal number of linearly independent vectors among* $\mathfrak{a}_1, \mathfrak{a}_2, ..., \mathfrak{a}_p$.

Theorem 23.12 *Any* k $(k \leq p)$ *linearly independent vectors*

$$\mathfrak{b}_1, \mathfrak{b}_2, ..., \mathfrak{b}_k$$

of L *can be extended to form a basis of* L *by suitably adjoining to them* $p-k$ *other vectors* $\mathfrak{a}_{k+1}, \mathfrak{a}_{k+2}, ..., \mathfrak{a}_p$.

Let L' and L'' be any two vector spaces in $\mathbb{R}_n$. By the **intersection** D of L' and L'' we mean the set of all those vectors belonging to both L' and L''.

If we form the totality S of all vectors of the form $\mathfrak{a}' + \mathfrak{a}''$, where $\mathfrak{a}'$ belongs to L' and $\mathfrak{a}''$ belongs to L''. S is called the **sum** of L' and L''.

Theorem 23.13 *If the dimensions of the vector spaces* L', L'', D, *and* S *are respectively* p', p'', d *and* s, *then*

$$p' + p'' = d + s.$$

24 Some Vector Theorems (3-dimensions)

$$\mathbf{B} \times \mathbf{C} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_x & B_y & B_z \\ C_x & X_y & C_z \end{bmatrix} = \mathbf{i}(B_yC_z - C_yB_z) + \mathbf{j}(B_zC_x - B_xC_z) + \mathbf{k}(B_xC_y - B_yC_x)$$

Theorem 24.1 (BAC-CAB)

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{a}(\mathbf{b} \cdot \mathbf{c})$$

Proof Given the expression for $\mathbf{B} \times \mathbf{C}$ we have

$$\begin{aligned} \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ (b_yc_z - b_zc_y) & (b_zc_x - b_xc_z) & (b_xc_y - b_yc_x) \end{bmatrix} \\ &= \mathbf{i}(a_y(b_xc_y - b_yc_x) - a_z(b_zc_x - b_xc_z)) + \\ &\quad \mathbf{j}(a_z(b_yc_z - c_yb_z) - a_x(b_xc_y - b_yc_x)) + \\ &\quad \mathbf{k}(a_x(b_zc_x - c_zb_x) - a_y(b_yc_z - b_zc_y)) \end{aligned}$$

Prove component-by-component[4]. The $\mathbf{i}^{th}$ component can be written as

$$b_x(a_yc_y + a_zc_z) - c_x(a_yb_y + a_zb_z) = b_x(a_xc_x + a_yc_y + a_zc_z) - c_x(a_xb_x + a_yb_y + a_zb_z) = b_x(\mathbf{a} \cdot \mathbf{c}) - c_x(\mathbf{a} \cdot \mathbf{b}).$$

The $\mathbf{j}^{th}$ component can be written as

$$b_y(a_zc_z + a_xc_x) - c_y(a_zb_z + a_xb_x) = b_y(a_xc_x + a_yc_y + a_zc_z) - c_y(a_xb_x + a_yb_y + a_zb_z) = b_y(\mathbf{a}\cdot\mathbf{c}) - c_y(\mathbf{a}\cdot\mathbf{b}).$$

The $\mathbf{k}^{th}$ component can be written as

$$b_z(a_xc_x + a_yc_y) - c_z(a_xb_x + a_yb_y) = b_z(a_xc_x + a_yc_y + a_zc_z) - c_z(a_xb_x + a_yb_y + a_zb_z) = b_z(\mathbf{a}\cdot\mathbf{c}) - c_z(\mathbf{a}\cdot\mathbf{b}).$$

Putting back the respective unit vectors and adding we find the vector sum to be equal to

$$\mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b}).$$

This demonstrates the proof that

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b}).$$

The second part of the theorem is proven in a similar manner:

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c} = -\mathbf{c}\times(\mathbf{a}\times\mathbf{b}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{c}),$$

using the first half of the proof along with the substitutions:

$$\mathbf{a}\to-\mathbf{c}\quad \mathbf{b}\to\mathbf{a}\quad \mathbf{c}\to\mathbf{b}$$

viz.

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{c} = -\mathbf{c}\times(\mathbf{a}\times\mathbf{b}) \to \mathbf{a}(-\mathbf{c}\cdot\mathbf{b}) - \mathbf{b}(-\mathbf{c}\cdot\mathbf{a}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{c}).$$

$\blacksquare$

Theorem 24.2

$$(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d}) = (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d}) - (\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c})$$

Proof

$$\mathbf{x}\cdot(\mathbf{c}\times\mathbf{d}) = (\mathbf{x}\times\mathbf{c})\cdot\mathbf{d}$$

Let $\mathbf{x} = \mathbf{a}\times\mathbf{b}$. Then

$$\begin{aligned}
(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{c}\times\mathbf{d}) &= \{(\mathbf{a}\times\mathbf{b})\times\mathbf{c}\}\cdot\mathbf{d} \\
&= \{\mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{c})\}\cdot\mathbf{d} \\
&= (\mathbf{a}\cdot\mathbf{c})(\mathbf{b}\cdot\mathbf{d}) - (\mathbf{a}\cdot\mathbf{d})(\mathbf{b}\cdot\mathbf{c}) \\
&= \det\begin{bmatrix}\mathbf{a}\cdot\mathbf{c} & \mathbf{a}\cdot\mathbf{d}\\ \mathbf{b}\cdot\mathbf{c} & \mathbf{b}\cdot\mathbf{d}\end{bmatrix}
\end{aligned}$$

Theorem 24.3

$$(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}\times\mathbf{d}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{c}\times\mathbf{d}) = \mathbf{c}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{d}) - \mathbf{d}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{c})$$

Proof

$$\mathbf{x}\times(\mathbf{c}\times\mathbf{d}) = \mathbf{c}(\mathbf{x}\cdot\mathbf{d}) - \mathbf{d}(\mathbf{x}\cdot\mathbf{c})$$

Let $\mathbf{x} = \mathbf{a}\times\mathbf{b}$. Then

$$\begin{aligned}
\mathbf{I}:\quad (\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) &= \mathbf{c}(\mathbf{a}\times\mathbf{b}\cdot\mathbf{d}) - \mathbf{d}(\mathbf{a}\times\mathbf{b}\cdot\mathbf{c}) \\
&= \mathbf{c}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{d}) - \mathbf{d}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}) \quad \blacksquare
\end{aligned}$$

Also

$$(\mathbf{a}\times\mathbf{b})\times\mathbf{y} = \mathbf{b}(\mathbf{a}\cdot\mathbf{y}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{y})$$

Let $\mathbf{y} = \mathbf{c}\times\mathbf{d}$. Then

$$\mathbf{II}: \quad (\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}\times\mathbf{d}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{c}\times\mathbf{d}) \qquad \blacksquare$$

Theorem 24.4 (Compound Vector Scalar Product)

$$(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a}) = (\mathbf{a}\cdot\mathbf{b}\times\mathbf{c})^2$$

Proof

$$\mathbf{x}\times(\mathbf{c}\times\mathbf{a}) = \mathbf{c}(\mathbf{x}\cdot\mathbf{a}) - \mathbf{a}(\mathbf{x}\cdot\mathbf{c})$$

Let $\mathbf{x} = \mathbf{b}\times\mathbf{c}$, then

$$\begin{aligned}(\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a}) &= \mathbf{c}(\mathbf{b}\times\mathbf{c}\cdot\mathbf{a}) - \mathbf{a}(\mathbf{b}\times\mathbf{c}\cdot\mathbf{c})\\ &= \mathbf{c}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}) - \mathbf{a}(\mathbf{b}\cdot\times\mathbf{c}\times\mathbf{c})\\ &= \mathbf{c}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{c}).\end{aligned}$$

Therefore

$$\begin{aligned}(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a}) &= (\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}(\mathbf{a}\cdot\mathbf{b}\times\mathbf{c})\\ &= (\mathbf{a}\times\mathbf{b}\cdot\mathbf{c})(\mathbf{a}\cdot\mathbf{b}\times\mathbf{c})\\ &= (\mathbf{a}\cdot\mathbf{b}\times\mathbf{c})^2 \qquad \blacksquare\end{aligned}$$

Definition The operator ∇ is given by

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}.$$

Definition The Divergence of a vector field is given by

$$\nabla\cdot\mathbf{a} = \frac{\partial}{\partial x}a_x + \frac{\partial}{\partial y}a_y + \frac{\partial}{\partial z}a_z.$$

Definition The Laplacian (Divergence of the Gradient) operator is given by

$$\nabla^2 = \nabla\cdot\nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Definition The Curl of a vector field is given by

$$\nabla\times\mathbf{a} = \det\begin{bmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \partial_x & \partial_y & \partial_z\\ a_x & a_y & a_z\end{bmatrix}$$

Theorem 24.5

$$\nabla\cdot(\mathbf{a}\times\mathbf{b}) = \mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b}).$$

Proof

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) = \det \begin{bmatrix} \partial_x & \partial_y & \partial_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{bmatrix}$$

$$\begin{aligned}
&= [\partial_x(a_y b_z - a_z b_y) - \partial_y(a_x b_z - a_z b_x) + \partial_z(a_x b_y - a_y b_x)] \\
&= b_z \partial_x a_y - b_y \partial_x a_z - b_z \partial_y a_x + b_x \partial_y a_z + b_y \partial_z a_x - b_x \partial_z a_y \\
&+ a_y \partial_x b_z - a_z \partial_x b_y - a_x \partial_y b_z + a_z \partial_y b_x + a_x \partial_z b_y - a_y \partial_z b_x \\
\\
&= [b_x(\partial_y a_z - \partial_z a_y) - b_y(\partial_x a_z - \partial_z a_x) + b_z(\partial_x a_y - \partial_y a_x)] \\
&- [a_x(\partial_y b_z - \partial_z b_y) - a_y(\partial_x b_z - \partial_z b_x) + a_z(\partial_x b_y - \partial_y b_x)] \\
&= \mathbf{b} \cdot (\nabla \times \mathbf{a}) - \mathbf{a} \cdot (\nabla \times \mathbf{b}) \qquad \blacksquare
\end{aligned}$$

Theorem 24.6

$$\nabla \times (\mathbf{a} \times \mathbf{b}) = (\mathbf{b} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{b} + (\nabla \cdot \mathbf{b})\mathbf{a} - (\nabla \cdot \mathbf{a})\mathbf{b}.$$

Theorem 24.7

$$\nabla \times (\nabla \times \mathbf{a}) = \nabla(\nabla \cdot \mathbf{a}) - (\nabla \cdot \nabla)\mathbf{a} = \nabla(\nabla \cdot \mathbf{a}) - \nabla^2 \mathbf{a}.$$

25 Reciprocal Vector Space (3-dimensions)

Take three non-coplanar vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ which form a right-handed system of vectors and form the vector scalar product whose magnitude is the volume, V, of the parallelapiped with edges given by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

$$V = \mathbf{a} \cdot \mathbf{b} \times \mathbf{c}.$$

The set of vectors $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ which are reciprocal to $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is given by

$$\begin{aligned}
\mathbf{a}' &= \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \\
\mathbf{b}' &= \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \\
\mathbf{c}' &= \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}
\end{aligned}$$

The reciprocal basis $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ satisfy the following identities

$$\begin{aligned}
\mathbf{a} \cdot \mathbf{a}' &= 1, \quad \mathbf{b} \cdot \mathbf{a}' = 0, \quad \mathbf{c} \cdot \mathbf{a}' = 0 \\
\mathbf{b} \cdot \mathbf{b}' &= 1, \quad \mathbf{a} \cdot \mathbf{b}' = 0, \quad \mathbf{c} \cdot \mathbf{b}' = 0 \\
\mathbf{c} \cdot \mathbf{c}' &= 1, \quad \mathbf{a} \cdot \mathbf{c}' = 0, \quad \mathbf{b} \cdot \mathbf{c}' = 0
\end{aligned}$$

Theorem 25.1 (Reciprocal Basis has Reciprocal Volume = 1/V)

$$\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}' = \frac{1}{V}$$

Proof

$$\begin{aligned}
\mathbf{a}' &= \frac{\mathbf{b} \times \mathbf{c}}{V} \\
\mathbf{b}' &= \frac{\mathbf{c} \times \mathbf{a}}{V} \\
\mathbf{c}' &= \frac{\mathbf{a} \times \mathbf{b}}{V}
\end{aligned}$$

$$\begin{aligned}
\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}' &= \frac{(\mathbf{b} \times \mathbf{c}) \cdot (\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})}{V^3} = \frac{(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})}{V^3} \\
&= \frac{(\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})^2}{V^3} = \frac{V^2}{V^3} = \frac{1}{V} \qquad \blacksquare
\end{aligned}$$

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26 Vandermonde Determinant

The Vandermonde Determinant for n variables $x = [x_1, x_2, \ldots, x_n]$ is given by

$$V_n(x) = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^n \\ 1 & x_2 & x_2^2 & x_2^3 & \ldots & x_2^n \\ 1 & x_3 & x_3^2 & x_3^3 & \ldots & x_2^n \\ \vdots \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^n \end{bmatrix}$$

Except for the first column, subtract x_1 times the previous column to the rest of the columns. Since a multiple of any column added to any other column does not change the determinant, we can write

$$\det V_n(x) = \det \begin{bmatrix} 1 & 0 & 0 & 0 & \ldots & 0 \\ 1 & x_2 - x_1 & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \ldots & x_2^{n-1}(x_2 - x_1) \\ 1 & x_3 - x_1 & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \ldots & x_3^{n-1}(x_3 - x_1) \\ \vdots \\ 1 & x_n - x_1 & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \ldots & x_n^{n-1}(x_n - x_1) \end{bmatrix}$$

Expanding the determinant using the elements in the first row we have

$$\det V_n(x) = \det \begin{bmatrix} x_2 - x_1 & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \ldots & x_2^{n-1}(x_2 - x_1) \\ x_3 - x_1 & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \ldots & x_3^{n-1}(x_3 - x_1) \\ \vdots \\ x_n - x_1 & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \ldots & x_n^{n-1}(x_n - x_1) \end{bmatrix}$$

Factoring out $x_2 - x_1$ from the first row, $x_3 - x_1$ from the second row, and so on until the last row we have

$$\det V_n(x_1, x_2, \ldots, x_n) = [\prod_{i=2}^{n}(x_i - x_1)] \det \begin{bmatrix} 1 & x_2 & x_2^2 & \ldots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \ldots & x_3^{n-1} \\ \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^{n-1} \end{bmatrix} = \prod_{i=2}^{n}(x_i - x_1) \ \det V_{n-1}(x_2, x_3, \ldots, x_n)$$

Proceeding in a similar way by subtracting x_2 times the previous column except for the first column we can continue

$$\det V_n(x_1, x_2, \ldots, x_n) = [\prod_{i=2}^{n}(x_i - x_1)]\,[\prod_{i=3}^{n}(x_i - x_2)]\,\det V_{n-2}(x_3, x_4, \ldots, x_n)$$

To arrive at

$$\det V_n(x_1, x_2, \ldots, x_n) = \prod_{\substack{1<j\\ i<j}}^{n}(x_j - x_i) = \prod_{1\le i<j\le n}(x_j - x_i)$$

27 Cayley-Hamilton Theorem

The characteristic equation for eigenvalues $0 = \det|A - \lambda I|$ can be expanded as a polynomial in λ

$$\phi(\lambda) = \det|A - \lambda I| = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \ldots + a_1\lambda + a_0, \quad a_n = (-1)^n.$$

And the characteristic equation can be written in the form $\phi(\lambda) = 0$.

If in any polynomial $f(\lambda) = \sum c_i\lambda^i$ we substitute the matrix A for λ and multiply the constant term c_0 by I, we obtain a matrix denoted by $f(A)$.

Theorem 27.1 *(Cayley-Hamilton) Any square matrix satisfies its characteristic equation.*

Proof Let $\phi(\lambda)$ be the characteristic determinant of A. We want to show that $\phi(A) = 0$. Since the elements of $A - \lambda I$ are linear functions of λ, and the elements of its adjoint [the transpose of the matrix of signed minors of $A - \lambda I$ – denoted by Adj-$(A - \lambda I)$] are (n-1)-rowed determinants, they are polynomials in λ of degree $\le n-1$. Let C stand for the Adj-$(A - \lambda I)$. If the element in the i-th row and j-th column of C is $\sum_k c_{ijk}\lambda^k$, then

$$C = \sum_{k=0}^{n-1} C_k\lambda^k, \quad C_k = (c_{ijk}) \quad (i, j = 1, 2, \ldots, n)$$

Using the relationship between the adjoint and the determinant, $\phi(\lambda) = \det|A - \lambda I|$, we can write

$$(A - \lambda I)C = \phi(\lambda)I.$$

This expression can be expanded as

$$A\sum_{k=0}^{n-1} C_k\lambda^k - \lambda\sum_{k=0}^{n-1} C_k\lambda^k = \sum_{k=0}^{n} a_k\lambda^k I$$

Equating the terms free of λ and the coefficients of $\lambda, \lambda^1, ..., \lambda^{n-1}, \lambda^n$ we get

$$\begin{aligned}
AC_0 &= a_0 I,\\
AC_1 - C_0 &= a_1 I,\\
AC_2 - C_1 &= a_2 I,\\
\ldots &\quad \ldots\\
AC_{n-1} - C_{n-2} &= a_{n-1} I,\\
-C_{n-1} &= a_n I.
\end{aligned}$$

Multiply these equations on the left by $I, A, A^2, ..., A^{n-1}, A^n$ respectively and add; we get

$$0 = a_0 I + a_1 A + a_2 A^2 + \ldots + a_{n-1}A^{n-1} + a_n A^n = \phi(A),$$

and we see that the matrix A satisfies its own characteristic equation. ∎

28 Sylvesters Criterion for Positive Definite Hermitian Matrices

Theorem 28.1 *(Sylvester's Criterion) An n-x-n Hermitian matrix M is positive-definite if and only if all the leading principal minors are positive.*

Proof Let M be an n x n Hermitian positive definite matrix. Then $0 \leq x^\dagger M x$ if x is not the zero vector. Let M_k be the k-th leading principal minor (the first k-rows and k-columns in the upper left hand corner of M). Select a non-zero vector x as follows

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_k \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \vec{x} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Then we have that $0 \leq x^\dagger M x = \vec{x}^\dagger M_k \vec{x}$. Therefore, since $\vec{x} \neq \vec{0}$ is otherwise arbitrary, then all the eigenvalues of M_k must be positive and hence $\det |M_k| > 0$

To prove the reverse implication, we use induction. The general form of an $(n+1)$-x-$(n+1)$ Hermitian matrix is

$$M_{n+1} = \begin{bmatrix} M_n & \vec{v} \\ \vec{v}^\dagger & d \end{bmatrix}$$

Denote the $(n+1)$ dimensional vector x as

$$x = \begin{bmatrix} \vec{x} \\ x_{n+1} \end{bmatrix}$$

Then

$$x^\dagger M_{n+1} x = \vec{x}^\dagger M_n \vec{x} + x_{n+1}\vec{x}^\dagger \vec{v} + \overline{x}_{n+1}\vec{v}^\dagger \vec{x} + d|x_{n+1}|^2$$

By completing the square, this last expression is equal to

$$\begin{aligned} &(\vec{x}^\dagger + \vec{v}^\dagger M_n^{-1}\overline{x}_{n+1}) M_n (\vec{x} + x_{n+1} M_n^{-1}\vec{v}) - |x_{n+1}|^2 \vec{v}^\dagger M_n^{-1}\vec{v} + d|x_{n+1}|^2 \\ &= (\vec{x} + \vec{c})^\dagger M_n (\vec{x} + \vec{c}) + |x_{n+1}|^2 (d - \vec{v}^\dagger M_n^{-1}\vec{v}), \end{aligned}$$

where $\vec{c} = x_{n+1} M_n^{-1}\vec{v}$. M_n^{-1} exists because the eigenvalues of M_n are all positive. The first term is positive by the induction hypothesis. Using this useful determinant expansion

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det |A| \det |D - CA^{-1}B|$$

We have that

$$\det |M_{n+1}| = \det |M_n| (d - \vec{v}^\dagger M_n^{-1}\vec{v}) > 0,$$

which implies $(d - \vec{v}^\dagger M_n^{-1}\vec{v}) > 0$. Hence $x^\dagger M_{n+1} x > 0$ (is also positive definite). ∎

29 Notes on Riemannian Hypersurfaces

Let us summarize some facts about Riemannian Geometry:

Tangent Vectors $\mathfrak{r}_j = \partial\mathfrak{r}/\partial x^j$, $\mathfrak{a} = a^i \mathfrak{r}_i$, second derivatives: $\mathfrak{r}_{ij} = \frac{\partial^2 \mathfrak{r}}{\partial x^i \partial x^j}$
Metric: $g_{ij} = \mathfrak{r}_i \cdot \mathfrak{r}_j = g_{ji}$ (symmetric).
Dot product: $\mathfrak{a} \cdot \mathfrak{b} = g_{ij} a^i b^j$.

Consider the determinant

$$g = \det|g_{ij}| = \begin{vmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ g_{21} & g_{22} & \cdots & g_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ g_{i1} & g_{i2} & \cdots & g_{in} \\ \cdots & \cdots & \cdots & \cdots \\ g_{n1} & g_{n2} & \cdots & g_{nn} \end{vmatrix} \tag{48}$$

We suppose, here and throughout, that g is not zero. Let Δ^{ij} be the cofactor of g_{ij} in this determinant, so that

$$g_{mr}\Delta^{ms} = g_{rm}\Delta^{sm} = \delta_r^s \, g,$$

which follows from the ordinary rules for developing a determinant.

Let us construct new quantities g^{ij}[6]. The values of the components of g^{ij} are equal to the cofactor of the g_{ij}, divided by the full determinant g,

$$g^{kl} = g^{-1}\Delta^{kl}. \tag{49}$$

Alternatively the cofactor of g_{ij} can be expressed in terms of g^{ij} as folllows

$$\Delta^{ij} = g\, g^{ij}. \tag{50}$$

g^{ij} satisfies the following equations:

$$g_{ik}g^{kl} = \delta_i^l \ \text{ inverse} \tag{51}$$

Now g_{ij} is symmetric and we can also show that g^{ij} is symmetric. Multiply Eq. (51) by $g_{ls}g^{ir}$. The left hand side becomes

$$g_{ik}g^{kl}g_{ls}g^{ir} = g_{ki}g^{ir}g^{kl}g_{sl} = \delta_k^r g^{kl}g_{sl} = g_{sl}g^{rl},$$

while the right hand side becomes

$$\delta_i^l g_{ls}g^{ir} = g_{is}g^{ir} = g_{si}g^{ir} = \delta_s^r.$$

Therefore we obtain

$$g_{sl}g^{rl} = \delta_s^r \ \text{ inverse} \tag{52}$$

Comparing Eqs (51) and (52), we find that

$$g^{kl} = g^{lk}$$

Since g_{ij} is symmetric, it is obvious that g^{ij} should be symmetric also.

Let us examine the implications of the above applied to surfaces embedded in $\mathbb{R}^n$. A *curve* is defined as the totality of points given by the equations

$$x^r = f^r(u) \quad (r = 1, 2, ..., n). \tag{53}$$

Here u is a parameter and $f^r(u)$ are n functions.

Next consider the totality of points given by

$$x^r = f^r(t^1, t^2, ..., t^m) \quad (r = 1, 2, ..., n), \tag{54}$$

where the t's are parameters and $m < n$. This set of points forms V_m, a subspace of V_n. According to the Implicit Function Theorem it is possible to eliminate the parameters from (54). As long as the system is differentiable enough and the the Jacobian of the first m equations does not vanish, we can find m functions $g^i(x^1, ..., x^m)$ such that

$$t^i = g^i(x^1, ..., x^m) \text{ for } i = 1, ..., m \tag{55}$$

which implies that the remaining functions $x^r, r = m + 1, ..., n$ can be written in terms of the first m coordinates $x^1, ..., x^m$ as follows:

$$x^r = f^r(x^1, ..., x^m) \text{ for } r = m + 1, ..., n \tag{56}$$

In the case of $m = n - 1$ there is only one such relation and so, elimination gives just one equation

$$\phi(x^1, x^2, ..., x^n) = x^n - f^n(x^1, ..., x^m) = 0. \tag{57}$$

In this case, V_{n-1} divides portions of V_n into two parts for which respectively ϕ is positive and negative. V_{n-1} is commonly referred to as a *hypersurface* in V_n.

Let V_n be a region of $\mathbb{R}^n$, bounded by a smooth (i.e. sufficiently differentiable) surface B_{n-1}, which is an $n - 1$ dimensional closed manifold. We take B_{n-1} as an example of a hypersurface V_{n-1}. We has suppose that B_{n-1} is parameterized by $n - 1$ independent parameters $t^\alpha(\alpha = 1, 2, ..., n - 1)$. Let the hypersurface B_{n-1} be defined implicitly by the equation (57).

On the surface of B_{n-1}, $x^i = x^i(t^\alpha)$, and therefore

$$\frac{\partial \phi}{\partial x^j}\frac{\partial x^j}{\partial t^\alpha} = 0. \tag{58}$$

Equation (58) may be regarded as $n - 1$ linear conditions upon the n quantities $\partial\phi/\partial x^j$. Since the matrix

$$\left[\frac{\partial x^j}{\partial t^a}\right]$$

is assumed to have its full rank value $n - 1$ on B_{n-1}, the mutual ratios of the partials $\partial\phi/\partial x^j$ are fully determined by (58).

For future reference let us define a set of determinants with respect to the change of variables between the x^i and the t^α

$$D_k = (-1)^{k-1} \det\left[\frac{\partial(x^1, ..., x^{k-1}, x^{k+1}, ..., x^n)}{\partial(t^1, ..., t^{n-1})}\right], \quad k = 1, 2, ..., n \tag{59}$$

The n determinants D_k satisfy the $n-1$ linear conditions

$$D_k \frac{\partial x^k}{\partial t^\alpha} = 0, \tag{60}$$

which can be established by noting that the left hand side can be written as an n-by-n determinant which has two rows the same and therefore vanishes. In view of (58, conditions (60) imply that the $\partial\phi/\partial x^k$ and D_k are proportional

$$\frac{\partial \phi}{\partial x^k} = \alpha D_k,$$

for some scalar factor of proportionality α. Hence the $\partial\phi/\partial x^k$ determine the unit normal to B_{n-1}, namely

$$n_k = \frac{1}{\sqrt{(\nabla\phi)^2}} \frac{\partial \phi}{\partial x^k} \tag{61}$$

$$n_k = \frac{D_k}{\sqrt{g^{ij} D_i D_j}} = \frac{D_k}{D}, \tag{62}$$

where

$$D^2 = g^{ij} D_i D_j = g^{ij} \frac{\partial \phi}{\partial x^i} \frac{\partial \phi}{\partial x^j} \alpha^{-2}.$$

30 Hypersurfaces and Concepts of Area

Consider an $n-1$ dimensional hypersurface, B_{n-1}, embedded in a Riemannian manifold M_n (which we assume to be sufficiently differentiable). Let $x^1, ..., x^n$ be local coordinates in M_n and suppose that B_{n-1} is represented locally in the form

$$x^j = x^j(t^1, ..., t^{n-1}), \quad j = 1, ..., n$$

The metric tensor of M_n is denoted by g_{ij}. and the metric tensor on B_{n-1} by $\overline{g}_{\alpha\beta}$. These two tensors are related by

$$\overline{g}_{\alpha\beta} = g_{ij} \frac{\partial x^i}{\partial t^\alpha} \frac{\partial x^j}{\partial t^\beta} = \frac{\partial x^i}{\partial t^\alpha} g_{ij} \frac{\partial x^j}{\partial t^\beta} \tag{63}$$

The element of area (or volume) in M_n is

$$dV = \sqrt{g}\, dx^1 ... dx^n, \tag{64}$$

where g is the determinant of the metric tensor in M_n. Similarly the element of area (volume) in the hypersurface B_{n-1} is given by

$$dS = \sqrt{\overline{g}}\, dt^1 ... dt^{n-1}. \tag{65}$$

We will use (63) to define the element of area in B_{n-1} based on expansion of the determinant of $\overline{g}$ in terms of products of the various minors in a manner similar to the Cauchy-Binet formula or the Laplace expansion of a determinant. Let S_n stand for the set of permutations of the numbers $\{1, 2...., n\}$ and let P represent one of the members of S_n.

$\overline{g}_{\alpha\beta}$ is an (n-1)-by-(n-1) matrix and we can consider its determinant.

$$\overline{g} = \det|\overline{g}| = \sum_{P \in S_{n-1}} \epsilon_{p_1 p_2 ... p_{n-1}} \overline{g}_{1p_1} \overline{g}_{2p_2} ... \overline{g}_{n-1 p_{n-1}} \tag{66}$$

We can think of the right hand side of (63) as representing the matrix product of three matrices.

$$\begin{aligned} J &= \left[\frac{\partial x^i}{\partial t^\alpha}\right] \text{ is } n\text{-by-}(n-1), \\ G &= \left[g\right] \text{ is } n\text{-by-}n \end{aligned}$$

and (63) can be written in the form:

$$\begin{aligned} \overline{g}_{\alpha\beta} &= \sum_{i,j=1}^{n} \left[\frac{\partial x^i}{\partial t^\alpha}\right]^T_{\alpha i} [g]_{ij} \left[\frac{\partial x^i}{\partial t^\beta}\right]_{j\beta} = \frac{\partial x^i}{\partial t^\alpha} g_{ij} \frac{\partial x^j}{\partial t^\beta} \text{ summation convention on } i \text{ and } j \\ \left[\overline{g}_{\alpha\beta}\right] &= J^T G J \end{aligned}$$

Inserting this computation into (66) we have $\overline{g}_{\alpha\beta}$ is an (n-1)-by-(n-1) matrix and we can consider its determinant. Let $\overline{g} = \det|\overline{g}_{\alpha\beta}|$.

$$\begin{aligned} \overline{g} &= \sum_{P\in S_{n-1}} \epsilon_{p_1 p_2 \ldots p_{n-1}} \sum_{\substack{i1,..,i_{n-1}=1 \\ k_1,\ldots,k_{n-1}=1}}^{n} (J^T_{1i_1} G_{i_1 k_1} J_{k_1 p_1})(J^T_{2i_2} G_{i_2 k_2} J_{k_2 p_2}) \ldots (J^T_{n-1 i_{n-1}} G_{i_{n-1} k_{n-1}} J_{k_{n-1} p_{n-1}}) \\ &= \sum_{P\in S_{n-1}} \epsilon_{p_1 p_2 \ldots p_{n-1}} \sum_{\substack{i1,..,i_{n-1}=1 \\ k_1,\ldots,k_{n-1}=1}}^{n} (J^T_{1i_1} \ldots J^T_{n-1 i_{n-1}})(G_{i_1 k_1} \ldots G_{i_{n-1} k_{n-1}})(J_{k1 p_1} \ldots J_{k_{n-1} p_{n-1}}) \\ &= \sum_{\substack{i1,..,i_{n-1}=1 \\ k_1,\ldots,k_{n-1}=1}}^{n} (J^T_{1i_1} \ldots J^T_{n-1 i_{n-1}})(G_{i_1 k_1} \ldots G_{i_{n-1} k_{n-1}}) \sum_{P\in S_{n-1}} \epsilon_{p_1 p_2 \ldots p_{n-1}} (J_{k1 p_1} \ldots J_{k_{n-1} p_{n-1}}) \end{aligned}$$

Examining the last term in the sum on the right hand side we find a sum over the $n-1$ set $\{p1, p2, ..., p_{n-1}\}$ for a particular choice of index values: $\{k1, ...k_{n-1}\}$

$$\sum_{P\in S_{n-1}} \epsilon_{p_1 p_2 \ldots p_{n-1}} (J_{k1 p_1} \ldots J_{k_{n-1} p_{n-1}}) = \epsilon_{k_1 k_2 \ldots k_{n-1}} |J_k| = \epsilon_{k_1 k_2 \ldots k_{n-1}} (-1)^{1+k} D_k \tag{67}$$

where the index k is not included in $\{k1, ...k_{n-1}\}$ and final result involves the respective determinant D_k.

At this point notice that the sum over the $n-1$ variables $k1, k2, ..., k_{n-1}$ involves summing over all values from 1 to n for each k_i, but that these values must be distinct. If $k_i = k_j$ for some pair i, j, then the sum is zero because of the presence of the permutation symbol $\epsilon_{k_1 k_2 \ldots k_{n-1}}$. This means that there are $n-1$ distinct k values chosen from the full set of n values, therefore one of the k_i values is to be omitted in this sum. Let k stand for this missing value and J_k stand for the matrix J with the k-th row deleted. The set of k_i values in any set is also propagated on to the G_{ik} matrices and the corresponding k-th column of G will not contribute this particular set of choices. There will be n such cases and therefore we insert a sum over $k = 1, 2, ..., n$ to account for each case in the following. The sum over the $i_1, ..., i_{n-1}$ indices follows that same way.

$$
\begin{aligned}
\overline{g} &= \sum_{k=1}^{n} \sum_{\substack{i1,..,i_{n-1}=1 \\ k_1,...,k_{n-1}=1,\neq k}}^{n} (J^T_{1i_1}...J^T_{n-1 i_{n-1}})(G_{i_1 k_1}...G_{i_{n-1}k_{n-1}})\epsilon_{k_1 k_2 ... k_{n-1}}(-1)^{1+k} D_k \\
&= \sum_{i,k=1}^{n} \sum_{i1,..,i_{n-1}=1,\neq i}^{n} (J^T_{1i_1}...J^T_{n-1 i_{n-1}})\epsilon_{i_1 i_2 ... i_{n-1}}(-1)^{i+k}\text{cofactor}(g_{ik})(-1)^{1+k} D_k \\
&= \sum_{i,k=1}^{n} \sum_{i1,..,i_{n-1}=1,\neq i}^{n} \epsilon_{i_1 i_2 ... i_{n-1}}(J^T_{1i_1}...J^T_{n-1 i_{n-1}})(-1)^{i+k}\Delta^{ik}(-1)^{1+k} D_k \\
&= \sum_{i,k=1}^{n} (-1)^{1+i} D_i (-1)^{i+k}\Delta^{ik}(-1)^{1+k} D_k \\
&= \sum_{i,k=1}^{n} D_i\, g\, g^{ik}\, D_k = g \sum_{i=1}^{n} D^k\, D_k = g\, D^2,
\end{aligned}
$$

where $g = |g_{ij}|$ as before and the sum is over the n values of i and k which correspond to the rows or columns which are crossed out to form the n minors each of dimension (n-1)-by-(n-1). Hence we find that

$$\overline{g} = g\, D^2.$$

We finally derive the relationship between the element of surface are on the hyperplane H and the metric of the n-dimensional Riemann M_n space that H is embedded inside.

$$dS = \sqrt{\overline{g}}\, dt^1 ... dt^{n-1} = \sqrt{g\, D^2}\, dt^1 ... dt^{n-1} \tag{68}$$

If the n-1 dimensional hypersurface H which is embedded in Cartesian $\mathbb{R}^n$ is given in parametric form as follows:

$$\mathfrak{r}(x) = \{x, f(x)\}$$

Then $g_{ij} = I_n$ (n-by-n identity matrix) and J is an n-by-$n-1$ matrix:

$$J = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \\ \frac{\partial f}{\partial x^1} & \frac{\partial f}{\partial x^2} & \dots & \frac{\partial f}{\partial x^{n-1}} \end{bmatrix} \tag{69}$$

$$J^T = \begin{bmatrix} 1 & 0 & \dots & 0 & \frac{\partial f}{\partial x^1} \\ 0 & 1 & \dots & 0 & \frac{\partial f}{\partial x^2} \\ \dots & \dots & \dots & \dots & \\ 0 & 0 & \dots & 1 & \frac{\partial f}{\partial x^n} \end{bmatrix} \tag{70}$$

Working out the minors, we find $D_n = 1$ (cross out the last row of J), and crossing out any of the other rows $i < n$ inserts a 0 in the i-th diagonal and shifts all the other rows $i+1, ..., n$ up one row. The resulting determinant is equal to $D_k = (-1)^{1+k} \frac{\partial f}{\partial x_k}$
Therefore we obtain

$$\overline{g} = \sum_{k=1}^{n} D_k = (\frac{\partial f}{\partial x^1})^2 + (\frac{\partial f}{\partial x^2})^2 + ... + (\frac{\partial f}{\partial x^n})^2 + 1 \tag{71}$$

And the resulting magnitude, given that $g_{ij} = I_n$ is $\sqrt{D^2} = \sqrt{1 + |\nabla f|^2}$ and

$$dS = \sqrt{1 + |\nabla f|^2}\; dt^1 ... dt^{n-1}. \tag{72}$$

31 Application to the Green Theorem

In Riemannian Geometry the divergence of a vector $\mathfrak{b}$ is given by

$$\nabla \cdot \mathfrak{b} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} (\sqrt{g}\, b^i) \text{ summation convention}$$

Let us form the integral of the divergence of $\mathfrak{b}$ over a region R in V_n bounded by the closed surface B_{n-1}

$$\int_R \nabla \cdot \mathfrak{b}\, dV = \int_R \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} (\sqrt{g}\, b^i)\, dV = \int_R \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} (\sqrt{g}\, b^i)\, \sqrt{g}\, dx^1 ... dx^n \tag{73}$$

Now let us integrate the differentiated terms over the range of the appropriate variable for each one. We obtain an integral over the boundary surface B_{n-1}: for the k term this is (k not summed)

$$\int_B \sqrt{g}\, b^k dx^1 ... dx^{k-1} dx^{k+1} ... dx^n = \int_B \sqrt{g}\, b^k D_k\, dt^1 ... dt^{n-1}, \tag{74}$$

where D_k is given as in (59) represents the transformation of the integration variables from the x^i to the coordinates as given on B_{n-1}.

From the above set of equations we see that

$$\begin{aligned} \int_{R_n} \nabla \cdot \mathfrak{b}\, dV_n &= \int_{B_{n-1}} b^k D_k \sqrt{g}\, dt^1 ... dt^{n-1} \\ &= \int_{B_{n-1}} b^k n_k \sqrt{g}\, D\; dt^1 ... dt^{n-1} \\ &= \int_{B_{n-1}} \mathfrak{b} \cdot \mathfrak{n}\, dS, \end{aligned}$$

where dS is given by (68).

Theorem 31.1 (Divergence Theorem)

$$\iiint_V A^k_{,k}\, dV = \iint_S A^k\, \nu_k\, dS$$

32 Extension and Stokes' Theorem

In a space of n dimensions, the **Generalized Kronecker delta** $\delta^{k_1\, k_2\, ...\, k_m}_{s_1\, s_2\, ...\, s_m}$ for any positive integer m is defined as

$\delta^{k_1\, k_2\, ... k_m}_{s_1\, s_2\, ... s_m} = +1$ if $k_1\, k_2\, ... k_m$ are distinct integers selected from the range $1, 2, ..., n$,
and if $s_1, s_2, ..., s_m$ is an *even* permutation of $k_1\, k_2\, ... k_m$.

$\delta^{k_1\, k_2\, ... k_m}_{s_1\, s_2\, ... s_m} = -1$ if $k_1\, k_2\, ... k_m$ are distinct integers selected from the range $1, 2, ..., n$,
and if $s_1, s_2, ..., s_m$ is an *odd* permutation of $k_1\, k_2\, ... k_m$.

$\delta^{k_1\,k_2\,\ldots k_m}_{s_1\,s_2\,\ldots s_m} = 0$ if any two $k_1\,k_2\,...k_m$ are, equal, or if any two $s_1, s_2, ..., s_m$ are equal, or if the set of numbers $\{k_1, k_2, ..., k_m\}$ differs, apart from order, from the set $\{s_1, s_2, ..., s_m\}$.

In a space of n dimensions, in which no metric is assigned, consider a subspace V_m of m dimensions $m \leq n$ defined by the parametric equations

$$x^k = x^k(y^1, y^2, ..., y^m),\ \ k = 1, 2, ..., n.$$

In this V_m consider a certain region of R_m. Let us divide R_m into cells by m families of surfaces

$$f^\alpha(y) = c^\alpha,\ \ \alpha = 1, 2, ..., m.$$

where c^α are constants, each taking on a number of discrete values and so forming a family of surfaces.

It is easy to attach a meaning to the statement that a point P of V_m lies in a certain cell, because P may be said to lie between two surfaces of the same family, say family α, if the expression $f^\alpha(y) - c^\alpha$ changes sign when we change c^α from the value belonging to one of these surfaces to the value belonging to the other. Here y refers to the parameter values at the point P.

Now pick one of the cells and let A be the point corresponding to one of the corners in the cell. Through A there passes one surface of each family. Let the corresponding constants be $c^1, c^2, ..., c^m$. Passing along the edge for which c^1 alone changes, we arrive at another corner B_1 for which the constants have values $c^1 + \Delta c^1, c^2, ..., c^m$. In passing from A to B_1 the parameters change from y^α at A to $y^\alpha + \Delta_1 y^\alpha$ at B_1 and the coordinates from x^k to $x^k + \Delta_1 x^k$. Similarly the parameter values and the coordinates of all corners $B_1, B_2, ..., B_m$ of the cell reached by traveling from A along an edge of the cell can be written as

$$\begin{aligned} &y^\alpha + \Delta_\beta y^\alpha, \\ &x^\alpha + \Delta_\beta x^k. \end{aligned}$$

and we form the determinants

$$\Delta^{k_1\,\ldots\,k_m} = \begin{vmatrix} \Delta_1 x^{k_1} & \Delta_1 x^{k_2} & \ldots & \Delta_1 x^{k_m} \\ \Delta_2 x^{k_1} & \Delta_2 x^{k_2} & \ldots & \Delta_2 x^{k_m} \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ \Delta_m x^{k_1} & \Delta_m x^{k_2} & \ldots & \Delta_m x^{k_m} \end{vmatrix} \tag{75}$$

Introducing the generalized Kronecker delta $\delta^{k_1\,k_2\,\ldots\,k_m}_{s_1\,s_2\,\ldots\,s_m}$, these determinants can be written compactly using the Summation Convention

$$\Delta^{k_1\,\ldots\,k_m} = \delta^{k_1\,k_2\,\ldots\,k_m}_{s_1\,s_2\,\ldots\,s_m}\,\Delta_1 x^{s_1}\,\Delta_2 x^{s_2}\ldots\Delta_m x^{s_m} \tag{76}$$

Now if the edges had been taken in a different order, then the various determinants might have opposite signs, but apart from this the determinants are independent of the order taken. Also if we have used a different corner to start from the determinants formed similarly would have different values. But since we are going to be working with infinitesimal, the differences will be of higher order than the determinants in (76). Using differential displacements, $d_\beta x^i$ (where β indexes the β-th edge of the cell) we have

$$d\tau_m^{k_1\,\ldots\,k_m} = \delta^{k_1\,k_2\,\ldots\,k_m}_{s_1\,s_2\,\ldots\,s_m}\,d_1 x^{s_1}\,d_2 x^{s_2}\ldots d_m x^{s_m} \tag{77}$$

The set of quantities $d\tau_m^{k_1\,\ldots\,k_m}$ is called the *extension* of the infinitesimal m-cell. No metrical concepts have entered into the definition of extension. Extension is the nearest concept closest to the idea of elementary volume one can define in a non-metrical space.

The edges of the infinitesimal m-cell can be written as

$$d_\beta x^k = \frac{\partial x^k}{\partial y^\alpha}\, d_\beta y^\alpha \tag{78}$$

Theorem 32.1 (Stoke's Theorem)

$$\oint_C A_p \frac{dx^p}{ds} ds = -\iint_S \epsilon^{pqr} A_{q,r}\, \nu_p\, dS,$$

where $\frac{dx^p}{ds}$ *is the unit tangent vector to the closed curve* C *and* ν_p *is the positive unit normal to the surface* S *which has* C *as its boundary.*

33 Applications In Mechanics

Definition Lie Algebras. Here $F = \mathbb{R}$ or $\mathbb{C}$. A Lie Algebra of F is a pair $(\mathfrak{g}, [\cdot,\cdot])$, where $\mathfrak{g}$ is a vector space over F and

$$[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

is an F-blinear map satisfying the following properties

$$\begin{aligned} &X, Y] = -[Y, X] \\ &[X, [Y, Z]]] + [Z, [X, Y]]] + [Y, [Z, X]]] = 0, \end{aligned}$$

The later is the Jacobi identity.

Definition Structure Constants: Given the elements X_a, X_b and X_c in a Lie Algebra we define the Structure Constraints c^c_{ab} to be the coefficients in the expansion of the Lie Bracket:

$$[X_a, X_b] = \sum_c c^c_{ab}\, X_c.$$

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